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Risk Management and
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MSc Climate Change,
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Group O

GROUP REPORT

Climate-Resilient Finance for Smallholders in Guatemala:
Bundling Crop Insurance with Innovative Production Technologies

Risk Management and Climate Change
Group O
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27/03/2026

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AI Disclaimer

We acknowledge the use of GenAI tools in the preparation of this report. The GenAI tools used were ChatGPT-5.1 (OpenAI), Gemini (Google DeepMind) and Perplexity AI. OpenAI, Google DeepMind and Perplexity.ai are the publishers of the tools. Their URLs are *https://chat.openai.com*, *https://gemini.google.com* and *https://www.perplexity.ai*.

We used these tools to clarify technical concepts, verify definitions, cross-check information and improve the clarity of some explanations during our writing process. All insights generated by GenAI tools were reviewed, verified and only used to guide our independent work.

We confirm that our scenario, analysis, calculations and final outputs presented in this project are our own work.

1 Introduction 5
2 Conceptual Rationale and Problem Framing 7
2.1 The economic logic of bundling insurance and input finance 7
2.2 Three linked mechanisms: risk transfer, credit enhancement, and adoption feedback 9
3 Demand-Side Challenges: Smallholder Farmers 12
3.1 Introduction 12
3.2 Feasibility & Eligibility 12
3.3 Demand-Side Basis Risk & Financial Instrument Trust 13
3.4 Technology Adoption Reluctance 14
3.5 Informal Structures & Contextual Importance 15
3.6 Closing Demand-Side Remarks 15
4 Supply-Side Challenges: Lenders and Insurers* 16
4.1 Insurer constraints: correlated risk, data, pricing, and reinsurance – TERESA 16
4.2 Lender constraints: portfolio risk, enforcement, and incentive compatibility – Max 17
4.3 Operational constraints: distribution, servicing, and claims settlement- Max 19
4.4 Regulatory Constraints: Strategic role of public policy and safety nets – Teresa 20
4.5 Implications for bundled product design – max 21
5 Evidence from Similar Experiments: Lessons Learned 22
5.1 Mechanisms for increasing uptake 22
5.1.1 Liquidity, timing and the 'bundling penalty' (Malawi, Ethiopia, Kenya, Uzbekistan) 22
5.1.3. Investment feedback and the resilience dividend (Tanzania and Mozambique, Northern Ghana) 23
5.2 Addressing basis risk and trust 24
5.2.1 Behavioural and cultural dimensions (Mexico, India) ..... **Error! Bookmark not defined.**
5.2.2 Data quality, index design and institutional responses to basis risk (Kenya) ..... **Error! Bookmark not defined.**
5.3 Institutional and operational requirements for scale ..... **Error! Bookmark not defined.**
5.3.1 Public provision (Mexico) ..... **Error! Bookmark not defined.**
5.3.2 Risk layering and public-private partnerships (Mexico, Mongolia)**Error! Bookmark not defined.**
5.3.3 Distribution, digital payments and trusted intermediaries (Kenya, India)**Error! Bookmark not defined.**
6 Policy Implications and Design Recommendations 26
6.1 Feasibility assessment 26
6.2 Product design choices/recommendations 27
6.3 Implementation Roadmap 29
6.4 Conditions for succes 30

6.5 Monitoring, evaluation, and learning 31
7 Conclusion..... 34

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Executive Summary

1 Introduction

Guatemala is one of the most climate-exposed countries in the world, yet it lacks the financial infrastructure to protect the farmers most at risk. However, it presents a unique landscape to explore _____

Ranked sixteenth globally for extreme weather vulnerability, the country generates over 80% of its GDP in zones exposed to drought, flooding, and hurricanes (Adil et al., 2026). The human face of this exposure is concentrated among its 1.3 million smallholder farmers, of whom roughly 86.5% depend entirely on rainfed agriculture, cultivate plots well below two hectares, and have access to formal credit in fewer than 8% of cases (Adil et al., 2026; FAO, 2018). The geography of vulnerability is not uniform. The Dry Corridor, a belt of semi-arid land running through the Pacific lowlands and interior valleys of Guatemala, El Salvador, Honduras, and Nicaragua, sits at the epicentre. Maize and bean farmers plantations in these areas face droughts that recur with increasing frequency, with the 1998, 2015, and 2019 El Niño events reducing yields by 30 to 40% in a single season and pushing entire communities into food insecurity (EM-DAT, 2024; WFP, 2022). Highland coffee growers face a different but equally precarious combination of erratic rainfall and frost risk, while sugarcane producers along the Pacific coast contend with hurricane exposure that can wipe out a season's output entirely. Climate change is tightening all three channels simultaneously, with projections pointing to more frequent and prolonged droughts, intensifying flood events, and a more unpredictable Atlantic hurricane season (Trocaire, 2025). What makes this situation particularly intractable is not climate exposure alone, but the combination of exposure and the near-total absence of instruments to manage it. Guatemala currently has only 97 meteorological stations across 109,000 km^2, a station density too sparse to calibrate reliable rainfall indices at the spatial resolution that smallholder agriculture requires (INSIVUMEH; World Bank DRIFCA, 2024). Formal agricultural insurance barely exists at scale: the Ministry of Agriculture, Livestock and Food (MAGA)'s Seguro Colectivo Paramétrico, the most significant domestic initiative, reached approximately 40,000 farmers in 2022, less than 3% of the smallholder population (MAGA, 2022). Most of those still without coverage rely on informal risk-sharing networks, a strategy that provides limited protection against the covariate shocks that define climate risk, since drought hits an entire community at once. Access to formal credit is equally constrained. Rural lending rates in informal markets exceed 20% per annum, partly because lenders cannot diversify their exposure when an entire region defaults simultaneously after a drought (Farrin and Miranda, 2015). The result is a financial system that effectively excludes the farmers who need it most. Our NGO seeks to introduce a bundled financial solution combining index-based crop insurance with input credit, enabling Guatemalan smallholders to invest in drought-tolerant maize and bean seeds, climate-resilient coffee varieties, climate-adapted sugarcane cultivars, and sustainable practices including agroforestry, while receiving a financial floor against the weather shocks most likely to derail their transition. The case for this bundle rests on the observation that the two barriers facing smallholders are not independent. Technology adoption requires upfront investment that informal credit markets price

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prohibitively, and even farmers who could access credit rationally, avoid doing so because a single failed harvest could render repayment impossible. Only an instrument that addresses both the liquidity constraint and the climate risk simultaneously can break this equilibrium. Yet the bundle's protective logic depends critically on the quality of its index triggers. Parametric insurance pays out on a measurable climatic indicator rather than observed on-farm losses, so those triggers must be calibrated on historical climate-yield relationships that are often sparse in Guatemala, especially at the spatial resolutions needed to reflect the country's microclimatic diversity (INSIVUMEH; World Bank DRIFCA, 2024). Calibration based on historical averages also performs poorly at the extremes, precisely where the most damaging events occur and where accurate threshold placement matters most. Thresholds therefore require regionally specific design and regular updating as climate conditions shift, adding an ongoing institutional burden that the NGO must plan for from the outset.

This report assesses the main challenges the NGO is likely to encounter in developing and scaling this solution. Two questions run through the analysis. The first is how to achieve meaningful uptake among a population that faces high barriers of affordability, trust, and financial literacy and whose existing informal risk networks actively compete with formal products. The second is how to manage basis risk, the structural gap between what the index measures and farmers' actual losses, which determines whether the product builds trust or destroys it at the moment it matters most.

Tia's proposed intro (220 words shorter):

Introduction

Ranked sixteenth globally for extreme weather vulnerability, Guatemala is one of the most vulnerable countries to climate change in the world (Adil et al., 2026). ~~the country~~The country generates over 80% of its GDP in zones exposed to drought, flooding, and hurricanes (Adil et al., 2026). ~~Guatemala is one of the most vulnerable countries to climate change in the world, yet~~Yet, it lacks the financial infrastructure required to protect farmers at high-risk to climate shocks. ~~Ranked sixteenth globally for extreme weather vulnerability, the country generates over 80% of its GDP in zones exposed to drought, flooding, and hurricanes (Adil et al., 2026).~~ This risk is concentrated among its 1.3 million smallholder farmers, with 86.5% depending entirely on rainfed agriculture, farm on land smaller than two hectares, and have minimal access to formal credit (fewer than 8% of cases) (Adil et al., 2026; FAO, 2018).

Climate risk is particularly acute in the 'Dry Corridor', a region classified as tropical dry forest or dry tropics, covering not only parts of Guatemala but also Costa Rica, Nicaragua, Honduras, El Salvador (FAO, no date). In this area, ~~where~~ recurrent droughts, such as those associated with El Niño events, have reduced maize and bean yields by up to 30-40% in a single season (EM-DAT, 2024). Other regions face different but equally significant risks, including frost in coffee-growing highlands and hurricane exposure along the Pacific coast (source?). Climate change is intensifying and exacerbating these patterns, increasing both the frequency and severity of shocks. Despite high exposure and vulnerability, effective financial insurance mechanisms and

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risk management measures remain almost absent in Guatemala. TThe Ministry of Agriculture, Livestock and Food (MAGA)'s Seguro Colectivo Paramétrico, the most significant domestic initiative, reached approximately 40,000 farmers in 2022, less than 3% of the smallholder population (MAGA, 2022), showing that formal agricultural insurance barely exists at scale. ~~Limited weather data~~Weather data collected through ~~(~~only 97 stations nationwide~~)~~ is not sufficient to calibrate reliable rainfall indices at the spatial resolution that smallholder agriculture requires, ~~constrains reliable index insurance,~~ while informal risk-sharing fails under covariate shocks such as drought (INSIVUMEH; World Bank DRIFCA, 2024). At the same time, access to credit is constrained, with informal lending rates exceeding 20% due to lenders' exposure to systemic default risk (Farrin and Miranda, 2015). The result is a financial system that effectively excludes smallholder farmers from investing in necessary climate-resilient technologies. Farmers face a fundamental trade-off: adopting improved seeds or practices requires upfront investment, but a single failed harvest can make repayment impossible. As a result, many remain locked into low-risk, low-productivity systems (Farrin and Miranda, 2015).

This report evaluates the challenges of ~~-~~introducing a bundled financial solution by a NGO, combining index-based crop insurance with input credit. The objective is to enable Guatemalan smallholder farmers to invest in climate resilient technologies such as, drought-tolerant maize, bean and coffee seeds, and agroforestry, whilst protecting them financially against the weather shocks that could be catastrophic to their livelihoods during the transition.

The viability of this approach depends upon two interlinked challenges. First, achieving meaningful uptake among a population that faces high barriers of affordability, trust, and financial literacy and whose existing informal risk networks actively compete with formal products. Second, managing basis risk, the structural gap between what the index-based payouts and farmers' actual losses, which determines whether the product builds trust and long-term participation. These challenges are closely linked: poor index performance reduces trust, which lowers uptake and undermines the longevity of the programme.

2 Conceptual Rationale and Problem Framing

2.1 The economic logic of bundling insurance and input finance

In the Guatemalan Dry Corridor, a maize or bean farmer approaching the start of a growing season faces a choice that is rational in its outcome but devastating in its aggregate effect. Drought-tolerant maize seeds from CIMMYT cost more than traditional varieties, and the additional inputs needed to benefit from them, fertilisers, better spacing, proper moisture management, push the upfront cost well beyond what a smallholder earning USD 300 to 400 per year can self-finance (Carter, Cheng and Sarris, 2016). Formal credit is available to fewer than 8% of this population (FAO, 2018). Informal lenders charge above 20% annually, partly because they cannot hedge against the event that a drought hits every borrower in the region at once (Farrin and Miranda, 2015; BANRURAL, 2022). Even a farmer who could borrow at those rates has good reason not to: a single season of 35% yield loss, the historical average for unimproved maize in an El Niño year (EM-DAT, 2024; FAOSTAT, 2024), generates income too low to repay the principal, let alone the interest. The rational response is to stay with traditional varieties, accept

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lower average yields, and avoid a debt burden that could permanently displace a subsistence household (Barnett, Barrett and Skees, 2008; FAO, 2018). This is the poverty trap in its concrete Guatemalan form, not a general theoretical construct but a specific equilibrium produced by the interaction of thin credit markets, high climate exposure, and the absence of risk transfer instruments (source?). Development economists have formalised this as a double constraint: the liquidity barrier that prevents investment, and the downside risk that makes investment irrational even if credit could be obtained (Farrin and Miranda, 2015). Neither instrument alone resolves it. Providing credit without insurance collapses at the first drought, as the farmer defaults and returns to the same position, now carrying debt. Providing insurance without credit leaves the investment unfinanced and the seeds unbought. The bundle targets both constraints simultaneously, making technology adoption individually viable rather than individually risky and collectively beneficial (Farrin and Miranda, 2015; Biffis et al., 2022). The relevant question for the NGO is therefore not whether the bundle is theoretically superior to either instrument in isolation, which the literature establishes clearly, but whether it can be designed and delivered in a way that is financially sustainable for insurers and lenders, accessible to the farmers who need it most, and reliable enough to build the trust on which uptake depends. It is worth noting that the literature is careful on one point: bundling generates its greatest benefit when both constraints are genuinely binding simultaneously (Carter, Cheng and Sarris, 2016). In contexts where credit access improves faster than climate risk management, or vice versa, the added value of the bundle over individual instruments shrinks. In the Guatemalan Dry Corridor, both constraints are acutely and simultaneously binding, which is precisely what makes this geography a compelling case.

Proposed alternative by Tia (200 words shorter):

2.1 The economic logic of bundling insurance and input finance

Smallholder farmers in Guatemala face two key constraints that limit investment in improved agricultural technologies: lack of access to finance and high exposure to climate risk (Farrin and Miranda, 2015). Adopting climate-resilient inputs, such as drought-tolerant seeds, requires upfront investment that exceeds the financial capacity of most smallholders (Carter, Cheng and Sarris, 2016). Formal credit is largely unavailable, and informal lending carries high interest rates reflecting systemic risk. Even when credit is accessible, farmers may rationally avoid borrowing, as climate shocks can render income insufficient to repay loans. This results in a poverty trap, where farmers remain in low-productivity systems despite the availability of higher-return technologies. The constraint is not simply financial but behavioural: the risk of catastrophic loss makes investment unattractive even when expected returns are positive. Credit and insurance individually do not resolve this problem. Credit alone increases exposure to default risk, while insurance alone does not enable investment (Carter, Cheng and Sarris, 2016). Bundling the two addresses both constraints simultaneously by enabling access to inputs while reducing downside risk (Farrin and Miranda, 2015; Biffis et al., 2022). The efficacy of this solution depends on context. Where either credit or risk constraints are weak, the benefits of bundling diminish (Carter, Cheng and Sarris, 2016). In the Guatemalan Dry Corridor, both constraints are strongly binding, making bundling particularly relevant.

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2.2 Three linked mechanisms: risk transfer, credit enhancement, and adoption feedback

The bundle works through three mechanisms, each of which is grounded in specific features of the Guatemalan context rather than generic financial theory.

The first is risk transfer. In the Dry Corridor, where drought years like 1998, 2009, and 2015 reduced maize yields by 30 to 40% in a single season (EM-DAT, 2024; FAOSTAT, 2024), the critical effect of insurance is not post-event compensation but pre-event behaviour change. A Dry Corridor farmer who knows that a certified drought will trigger a payout sufficient to cover loan repayment can rationally commit to drought-tolerant maize seeds at the start of the season without maintaining a low-input fallback strategy. This is what Carter, Cheng and Sarris (2016) identify as the ex-ante adoption effect: insurance does not merely protect income, it changes what farmers are willing to invest. The WFP and Aseguradora Rural pilot in Alta Verapaz and Chiquimula, which reached over 14,000 farmers with a USD 23 premium and up to USD 309 payout, provides direct evidence that parametric products calibrated to Dry Corridor rainfall patterns can be delivered at meaningful scale and generate this kind of behavioural shift (WFP, 2022). The trade-off is basis risk. Index insurance pays on a measured indicator, not on observed farm losses, and in a country with only 97 meteorological stations covering 109,000 km^2 (INSIVUMEH; World Bank DRIFCA, 2024), the spatial mismatch between what the index records and what any individual farmer experiences can be substantial. This mismatch originates in the design of the trigger itself: index triggers must be calibrated on historical climate-yield relationships, yet in Guatemala that data is sparse, particularly at the high spatial resolutions needed to capture the country's microclimatic diversity. Calibration using historical averages performs poorly at the extremes, precisely where the most damaging events occur and where accurate threshold placement is most critical. Thresholds therefore require regionally specific design and regular updating, adding an ongoing institutional burden to the NGO's implementation plan (World Bank DRIFCA, 2024; Biffis and Chavez, 2017). A farmer who suffers a 35% yield loss but whose nearest station recorded sufficient rainfall to keep the index above the trigger receives no payout and still owes loan repayments, a situation that is actively worse than having no insurance at all. Managing this trade-off between the cost efficiency of parametric products and the trust-destroying potential of basis risk events is one of the two central design challenges this report addresses.

The second mechanism is credit enhancement. Guatemala's rural lending rates exceed 20% per annum in informal markets not because lenders are exploitative but because a drought year creates a portfolio-wide default event: every maize farmer in a department fails to repay simultaneously, leaving the lender with no diversification (Farrin and Miranda, 2015). When index insurance covers this covariate tail risk, the lender's expected loss in a drought year falls materially, and the risk premium embedded in the interest rate can compress (Farrin and Miranda, 2015). The practical implication for Guatemala is significant: at 20% annual interest, a USD 350 input loan costs USD 70 per year, roughly 20% of a smallholder's gross income. At 16% under the bundle, the same loan costs USD 56, freeing up income that can fund further investment or absorb production shocks (Farrin and Miranda, 2015; BANRURAL, 2022). The

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2. Credit Enhancement

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bundle therefore generates a financial benefit not only for the farmer directly but for the credit market structure in which both farmer and lender operate. The trade-off here is eligibility. Accessing a bundled loan still typically requires some collateral signal, whether a land title, a credit history, or membership in a recognised cooperative. In Guatemala, where indigenous and smallholder farmers frequently cultivate communally held or informally occupied land with no formal title, this gateway requirement can exclude precisely the most vulnerable, those the NGO most wants to reach (Carter, Cheng and Sarris, 2016; Biffis et al., 2022).

The third mechanism is adoption feedback. As farmers in the Dry Corridor shift from traditional to drought-tolerant maize, or from conventional to shade-grown coffee varieties in the highlands, their yield profile gradually becomes less correlated with rainfall variability. A portfolio of insured farmers in their third year of resilient seed adoption generates smaller and less synchronised losses than the same portfolio in year one (Biffis et al., 2022). This means the insurer's loss ratio improves over time, creating the actuarial space to reduce premiums, which in turn lowers the cost of the bundle and widens the pool of farmers for whom it is affordable. The feedback loop, if it operates as theory predicts, makes the bundle progressively more self-sustaining. The trade-off is that successful adoption disrupts the very data on which pricing was based. Actuarial models calibrated on historical yield-rainfall relationships for traditional varieties become systematically miscalibrated as those varieties are replaced. Insurers who fail to update their models will underprice risk in normal years and overprice it in drought years, eroding both financial sustainability and farmer trust (Biffis et al., 2022; Biffis and Chavez, 2017).

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2.2.1 Risk transfer: enabling ex-ante investment

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In drought-prone regions such as Guatemala's Dry Corridor, the primary value of insurance lies in changing behaviour before shocks occur, rather than compensating losses after the fact.

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Historical droughts (e.g. 1998, 2009, 2015) have reduced maize yields by 30-40%, making investment in improved inputs highly risky (EM-DAT, 2024; FAOSTAT, 2024). By guaranteeing a payout in the event of a drought, insurance enables farmers to adopt higher-risk, higher-return technologies, such as drought-tolerant seeds, without maintaining low-input fallback strategies. This "ex-ante adoption effect" is well established in the literature (Carter, Cheng and Sarris, 2016). Evidence from a WFP-Aseguradora Rural pilot in Alta Verapaz and Chiquimula, which reached over 14,000 farmers with low premiums (USD 23) and meaningful payouts (up to USD 309), demonstrates that parametric insurance can be delivered at scale and influence investment decisions (WFP, 2022).

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This mechanism is constrained by basis risk. Index insurance pays based on a climatic indicator rather than realised farm losses, and in Guatemala, sparse meteorological infrastructure (97 stations nationwide) limits accurate calibration (INSIVUMEH; World Bank DRIFCA, 2024). As a result, farmers may experience substantial losses without triggering a payout. This "false negative" scenario where a farmer loses roughly 35% of output but still repays the loan, can undermine trust and reduce future uptake. Mana

ging this trade-off between cost-efficient parametric insurance and trust-damaging basis risk is a central design challenge.s
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2.2.2 Credit enhancement: expanding access to finance

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Insurance also improves access to credit by reducing lenders’ exposure to covariate risk. In Guatemala, drought can trigger widespread default, making agricultural lending inherently risky and leading to high interest rates (often above 20%) (Farrin and Miranda, 2015).
By covering systemic weather risk, index insurance reduces expected losses for lenders and can lower the risk premium embedded in loan pricing. For example, reducing interest rates from 20% to 16% on a USD 350 input loan lowers annual repayment costs from USD 70 to USD 56, a meaningful saving relative to smallholder income (BANRURAL, 2022). This mechanism expands credit supply and enables farmers to finance productivity-enhancing investments.

Despite improved credit conditions, access remains ~~constrained~~inhibited by eligibility requirements such as collateral, credit history, or cooperative membership. In Guatemala, many smallholders operate on informal or communal land without formal titles, limiting their ability to access bundled finance (Carter, Cheng and Sarris, 2016).
As a result, while insurance can make lending more viable by reducing risk for lenders, it does not automatically ensure inclusion of the most vulnerable farmers.

2.2.3 Adoption feedback~~: improving long-term sustainability~~

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Over time, the adoption of climate-resilient technologies can reduce farmers’ exposure to weather risk. As production becomes less sensitive to rainfall variability, insurance losses decline, improving the insurer’s loss ratio and creating scope for lower premiums (Biffis et al., 2022). This generates a positive feedback loop: adoption reduces risk, lower risk reduces insurance costs and lower costs increase affordability and uptake. If ~~sustained~~continued, this dynamic can make the bundled product progressively more self-sustaining.

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However, this process also undermines the data used to price insurance. As farmers shift from traditional to improved technologies, historical climate-yield relationships become less accurate. Without regular recalibration, insurers risk mispricing contracts, underestimating risk in some conditions and overestimating it in others and thereby threatening both financial sustainability and farmer trust (Biffis et al., 2022; Biffis and Chavez, 2017).

3 Demand-Side Challenges: Smallholder Farmers

3.1 Introduction

As a core stakeholder in this context, Guatemalan smallholder farmers face several types of demand-side challenges, limiting their ability to uptake bundled financing options and adopt innovative agricultural technologies. This is due to limited farmer understanding, trust and access to these finance bundles, crucially underpinned by the existing agro-finance structures and location-specific context. These barriers often interface and compound, severely limiting the agency of NGOs and governments in implementing and assisting these solutions, respectively.

3.2 Feasibility & Eligibility

Successful bundled solutions are highly dependent on their accessibility to those who require them most, that being poor family farmers that are vulnerable to adverse climatic impacts, 70% of which live below the poverty line in Northern Central America (World Bank, 2024). These financing solutions are able to derisk credit channels for these farmers via their index insurance lever, however this has been demonstrated with Tanzanian maize farmers to reduce, not necessarily eliminate the need for additional informational and financial assurances, such as credit histories and collateral assets (Biffis et al., 2022). If these novel bundles are unreliable or not sufficiently trusted by lenders due to their nascent nature, this insurance may be inadequate in its own right to unlock credit. Further requirements push the risk back onto the farmers critically when their financial resources are lowest, further stripping them of their assets and inventory after failed repayment.

Beyond these regressive impacts, insufficient insurance backing could also result in social exclusion, as with requested credit history. Small holder farmers, practicing traditional methods may have previous difficulties with credit tools or lack prior access to these instruments, providing the lenders with little alternative but to deny them access to innovative financing opportunities. Furthermore, when identifying insurance payees, land titles, ownership and boundaries are also usually required as to ensure potential compensation is sufficient but not overdistributed. Traditional agricultural networks and indigenous communities that commonly practice on communally occupied land, sometimes with no formal ownership complicates this process, potentially increasing the excluded population (World Bank, 2016).
A form of indirect exclusion from these initiatives may be through poor financial literacy and understanding, especially for these smallholder farmers with little exposure to complex financial instruments. Index insurance already complicates this interaction beyond simpler and more commonly used indemnity schemes and when these are overlaid with various credit tools with

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financial and physical disbursement mechanisms, these bundles can be difficult to understand and trust, especially if unobservable. With under 40% of adults in Northern Central America being financially literate and rural citizens being less likely to be financially included, NGOs looking to integrate into Guatemala will face complicated stakeholder interactions, requiring educational assistance schemes (World Bank, 2024).

Finally, individual viability is essential for these farmers as they need to be financially able to repay these loans. Even if the insurance is able to derisk the loans, this does not directly mean that they are unlocked for the farmers themselves as interest rates that remain too high, or are perceived to be, will hinder bundle and technological adoption rates given the excessive individual risk of failed repayment. Existing insurance mechanisms integrated in the Guatemala MAGA Seguro Colectivo Paramétrico Program are strong foundations on which to add credit channels, but these must be financially feasible for farmers.

If successful in early year implementation, trust between smallholder farmers, lenders and insurers can develop, as technological innovation increase yields and payouts compensate losses when required. This is however dependent on effective assimilation of financial tools at affordable levels, preventing them from being viewed as higher-interest borrowing mechanisms.

3.3 Demand-Side Basis Risk & Financial Instrument Trust

These social exclusion challenges are further exacerbated by the farmer uncertainty derived from financial basis risk. This can be applicable to stakeholders on the demand and supply-side of these operations, and they are both fundamentally dependent on the effective threshold design in these index insurance tools.

Thresholds that overestimate the required parameter level, for instance excessive precipitation or hurricane windspeeds, can result in scenarios where crops are damaged, but the insurance payout is not triggered. When combined with debt obligations for agricultural technologies, lost harvests and associated revenues are felt even more. This is highlighted as being a current issue facing smallholder farmers as the Guatemala MAGA Seguro Colectivo Paramétrico Program that monitors and sets thresholds for climate parameters currently operates on insufficient spatial resolutions of 27km by 27km. This is found to potentially result in inconsistent observations and unreliable monitoring in agricultural regions with complex topography, where impacts may be felt significantly but not captured by these tools (World Bank, 2024).

These poor thresholds result in worsened standards of living through lost livelihoods or through collateral damage due to credit defaulting. Furthermore, this defaulting on loans will increase future lender risk and therefore future interest rates, pricing out more of the farmers who need to benefit from these schemes the most. This is likely to occur even in the case of partial collateral reimbursement, which will negatively impact both the currently engaged farmers and discourage future potential stakeholders. Some farmers may have physical inventory or financial reserves to absorb the lost harvest revenue, however they still will bear negative economic consequences of investing in insurance that doesn't cover their losses, additionally making them less capable of investing in other agro-tech solutions in the future.

In addition to these core demand-side basis risk scenarios, insufficient payout compensation could result in similar worsening of farmers livelihoods. This shows how a misalignment between harvest revenue losses and insurance payouts, even in the case of effective threshold setting can still cause active harm to farmers operations. After implementation, peer effects also becomes an important factor. Observed negative experiences can spread through tight-knit communities, magnifying lowered adoption rates as long-standing opposition to financial and technological interventions develop.

Therefore, with such downside individual risks, trust in and engagement with these instruments remains low, as farmers opt to continue known and observed traditional practices instead, avoiding unnecessary losses.

3.4 Technology Adoption Reluctance

In addition to financial risk, farmers also experience behavioural barriers with uncertain technologies resulting in a low willingness to adopt unfamiliar practices.

The milpa farming practice of intercropping maize, beans and squash is an example of dominant traditional farming practices that have been passed on through generations and engrained in communities culture (Lopez-Ridaura et al., 2021). With its proven success and limited historical records of anomalous failed harvests, farmers may not fully comprehend the requirement for or the effectiveness other technological innovations that combat climate shocks. Compounding this, they may also be unaware of the increasingly common and extreme nature of these climate events.

This resultant loss aversion is also underpinned by present bias as farmers may discount future financial benefits significantly when compared to the immediate up-front costs, even if they are offset by credit channels, which would instead produce short-term repayment requirements. A relevant example of this is through a misalignment between early-stage costs and longer-term benefits with innovative practices like agroforestry. Tree growing durations and associated output benefits, including better soil structure, lower nutrient stripping and enhanced biodiversity, are often experienced in the longer-term and over multiple harvest periods. This means that present bias may magnify short-term investment costs and hinder technology adoption due to higher perceived risks of these benefits not materialising sufficiently, as there is more time over which these climate shocks can impact farm production. This results in lower perceived aggregate financial benefits for the farmers to capitalise on and with high technological scepticism, adoption also falls.

Similarly, this loss aversion may also be compounded by extreme climate shocks that damage harvest yields regardless of the technology implementation, framing these investments are unnecessary, complex and ultimately financially harmful. This technological risk adds to financial uncertainty with farmers potentially being overloaded with new concepts, products and offerings from external stakeholders who may fail to accommodate to context specific requirements (Mobarak & Rosenzweig, 2013). However, if successful in early years, which, as discussed is most likely to be in the present rather than the future, positive peer effects could accelerate adoption beyond an initial, cautious phase in, as farmers learn from their own and their communities' implementation as seen with more resilient coffee types following the 2013 coffee rust crisis (USDA Foreign Agricultural Service, 2016)

3.5 Informal Structures & Contextual Importance

An important intervening factor for the NGOs bundled finance solutions is the pre-existing context. These insurance and credit mechanisms are not able to penetrate the market freely as they don't enter into a vacuum.

One key element here are antecedent informal insurance networks. These are non-binding or contractual in nature but are instead built on social trust and long-term relationships between stakeholders on the same level; that being farmers. When adverse climate impacts reduce harvest yields and revenues, farmers aid their community through transfers of cash, inventory, labour or assets, contributing to individual and community resilience. Whilst these networks may lack guaranteed assurance, there is evidence that they do reduce demand for formal insurance schemes, including the proposed indexed instruments, as the farmers view these channels are sufficient to cope with risk (Mobarak & Rosenzweig, 2013).

Not only does informal insurance result in reduced demand for its formal counterparts, but they may also bear covariate risk. This is evident when formal insurance mechanisms don't pay out as parametric thresholds are not triggered, but crops are damaged and revenues lost for farmers engaged in both formal and informal schemes.

Alongside this lack of diversification in insurance risk, social dynamics may complicate formal instrument adoption. In the probable scenario that not all farmers switch from informal structures to these formal bundles, for reasons explained in section 3.2, the observation of members opting out of community support schemes could cause social divides and reduce community trust and informal insurance effectiveness. Local insurers and other financial operatives are keen for their existing structures to be integrated into new platforms, rather than a new macro-scale solution replacing them (World Bank, 2024). NGOs will want to work with local stakeholders to avoid potentially disruptive consequences and to better their relationships with local communities which will aid their social licence to operate and ultimately effective bundle and technology adoption. This is especially relevant given they are working with lenders and insurers who are perceived as external stakeholders attempting to penetrate existing markets. Failure to adjust operations and effectively integrate these instruments and practices into existing cultures is a core reason for slow agroforestry adoption in global and Central American coffee plots (Wienhold & Goulao, 2023). Fostering strong relations will help them develop effective operations as exemplified by informal insurance networks, materialising their expected improvements in crop yields, protection and farmer livelihoods.

Finally, any specific acute climate shocks immediately preceding the bundles' introduction will likely have a negative impact on adoption as farmers will be recouping losses from informal networks or will have lower financial reserves, potentially translating to higher perceived risks of technology engagement.

3.6 Closing Demand-Side Remarks

Evidently there are several categories of pertinent demand-side factors that limit farmers willingness and ability to integrate into this planned NGO scheme, with them being modulated by climate change through its systemic reach. As climate change increases the probability of extreme events, static index insurance payouts become more likely, assuming effective thresholds are set, but these damaging basis risk scenarios also increase in probability further

exacerbating farmer uncertainty. Critically, as outlined in the introduction, thresholds will require regular and consistent updating as climatic event probabilities evolve, but also as community resilience and social development occurs. The extent to which this is done effectively will have systemic impacts on all stakeholders involved, but the magnitude of any negative effects are likely to disproportionately hit farmers as the poorest, most vulnerable involved.

These challenges are also not independent, as they interact and compound for example with farmers that are technological and financially uncertain, already embedded in informal support networks, and therefore may prefer to remain exposed to damaging climate risks without accruing financial liabilities with little guaranteed positive returns. Experiences like this explain why insurance lacks market penetration accounting for only 1.9% of the national GDP in Guatemala, even before breaking this down into indemnity and index varieties (World Bank, 2024). This will therefore require bolstering by strong public policy support as seen with previous national agricultural policies aimed at increasing commercial development and financial inclusion of smallholder farmers to provide future technical assistance avenues to address these barriers in an integrated way, whilst critically cultivating positive local stakeholder interactions and relationships. (World Bank 2024; Mazviona & Sølvsten, 2025) Can index insurance keep up with climate change? Rethinking historical data models. Front. Clim. 7:1649540. doi: 10.3389/fclim.2025.1649540). Evidence from Tanzania (Biffis et al., 2022) and India (Mobarak & Rosenzweig, 2013) show that these financial bundles are achievable and can generate positive impacts, but they require sustained investment in index quality, dedicated community engagement, and product contextualisation.

4 Supply-Side Challenges: Lenders and Insurers*

4.1 Insurer constraints: correlated risk, data, pricing, and reinsurance –

Correlated Risk and Lack of Diversification

The proposed bundled finance solution presents unique challenges for traditional insurers adapting their products and services to the ongoing climate crisis and for those trying to understand the correlated risks of providing the novel bundled insurance solution proposed by the NGO. While traditional insurance mechanisms rely on the diversification of risk, climate risks are inherently systemic and correlated. Extreme weather events, such as droughts or flooding, tend to impact entire agricultural areas and can lead to large, aggregated losses. Smallholder farmers in a region will all suffer losses at the same time, so local insurers cannot easily diversify their exposure and may face insolvency risk if a major climate event occurs (World Bank, 2024).

This is indicated in the model by the fact that a higher-quality index with greater probability of an extreme weather event occurring (0.85) produces a higher "default probability" than the lower-quality S1 index and creates greater losses for the insurer during bad years. Because the farm yields correlate closely with the index, the insurer faces "synchronized losses" instead of many small, uncorrelated claims as the high-quality index triggers massive, simultaneous payouts across the entire region during a drought year. This requires the insurer to either have more capital available or to buy more expensive reinsurance to avoid insolvency. As hazards intensify, insurers may therefore pull back from high-risk markets altogether.

Reinsurance Market

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As local insurers in developing markets have limited reserves and capacity to manage highly correlated risk, there has been significant dependence on transferring aggregated risk to international reinsurers. Local insurers will typically front and cede 100% of the risk to the small number of international reinsurers willing to underwrite this class of business (World Bank, 2024). However, In Guatemala and the broader North Central American (NCA) region, early-stage agricultural insurance programs operate at a small scale and may struggle to generate the premium volumes needed to attract support from international reinsurers. As such, local insurers face increasing pressure to retain more risk, which they are ill-equipped to do. Furthermore, reinsurers rely on catastrophe models that struggle to adapt to rapidly changing climate baselines and correlated extremes. When actual losses diverge from modeled expectations, it generates volatility in pricing and capacity, and reinsurers tend to withdraw from markets when cumulative losses exceed their risk tolerances.

Basis Risk: Technology, Data and Cost Concerns

Insurers require vast amounts of high-quality, long-term meteorological and agronomic data to accurately price premiums and payouts. However, Guatemala only has 97 meteorological stations, many of which are still not fully automatic or connected to the national system (OECD, 2021). This is particularly relevant for Parametric insurance such as weather index insurance, where greater access to meteorological data allows insurers to price risk more accurately. This data limitation could therefore lead to basis risk. Furthermore, farmers are reluctant to pay market premiums where there is the risk of payouts being less than what is needed or where they have observed basis risk occur with peers. In some cases, when a small number of farmers bought insurance, financial institutions could not earn enough profit to make continued provision of insurance a worthwhile investment for them (J-PAL 2024).

As shown in the model, If the basis risk causes a false negative, farmers lose trust and disadopt the product, mathematically demonstrating how basis risk destroys the insurer's client base and could prevent the innovative product from reaching scale across the population. An approach to mitigating this risk could be the implementation of on-the-ground audits to verify areas where geo-spatial misalignment occurs. Modeled in Scenario 5, the simulation shows that implementing on-the-ground audits at a cost of $850 would successfully reduce basis-risk dis-adoption by 25% and boosts farmer trust recovery by 50%. However, the operational costs insurers would bear to physically verify losses would effectively eliminate the primary cost-saving advantage of parametric index insurance.

4.2 Lender constraints: portfolio risk, enforcement, and incentive compatibility – Max

This subsection examines how climate related portfolio risk, weak enforcement and the structure of incentives constrain the willingness of lenders to finance Guatemalan smallholders' adoption of new agricultural technologies. The focus is on rural cooperatives, microfinance institutions and public banks, which together provide the bulk of formal credit to smallholder farmers in the country (World Bank, 2023).

Commented [PM37]: In Guatemala, about 92 percent of farms are peasant owned smallholdings. Rural lenders, such as cooperatives, microfinance institutions and public banks lend to a very similar group of borrowers, mainly cultivating maize, beans and coffee. When a drought, flood or other climate related hazard occurs, many of these farmers face repayment difficulties at the same time, which can place severe stress on the lenders loan portfolio. A lot of the smallholders lack formal land titles or valuable collateral, and legal enforcement procedures are slow and costly, so lenders often response by charging high interest rates, shortening loan maturities or refusing to finance riskier farmers and new technologies altogether. Within this relationship, lenders do deploy incentive schemes, but these are usually simple and not explicitly linked to climate risk. As an example, they may offer slightly better terms to borrowers with strong repayment histories, membership in trusted borrowing groups or adoption of recommended agronomic practices that the lender believes improve repayment prospects. However, because lenders themselves remain highly exposed to climate related losses and continue to operate with limited effective collateral their capacity to offer more generous and climate sensitive incentives is constrained by their own risk exposure and capital position.
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Version A

Portfolio Risk:

From a portfolio perspective, these institutions exhibit a high concentration of exposure to climate sensitive activities. Most of their agricultural borrowers are smallholders cultivating maize, beans and coffee, often in areas that share similar rainfall patterns and soil conditions (FAO, 2018; Root Capital, 2014). This lack of diversification by crop, geography and client type implies that climate related hazards such as droughts, which have become more frequent in Guatemala's Dry Corridor, induce strongly correlated income shocks across borrowers (World Bank, 2024). When a severe drought occurs, a large share of farmers experience yield losses and liquidity shortfalls at the same time, leading to simultaneous repayment difficulties and sharp increases in lenders' portfolio at risk (World Bank, 2024). Index and parametric insurance products have been introduced to mitigate such shocks, but rising basis risk under climate change implies that payouts may not be triggered precisely in the seasons when individual farmers and lenders suffer the largest losses (Carter et al., 2016). As a result, lenders cannot fully rely on insurance as an effective hedge for their loan books and must continue to manage the impact of climate shocks on credit risk on their own balance sheets. This elevated residual exposure leads lenders to be more cautious in extending credit to smallholders, particularly for investments in untested technologies whose returns remain uncertain under evolving climatic conditions.

Enforcement:

These portfolio risks are further amplified by the enforcement environment. A large proportion of smallholders in Guatemala operate under informal or customary land tenure arrangements and have limited formal documentation of property rights (Méthot, 2018; FAO, 2018). Tangible assets that can serve as collateral are scarce, and even when land or equipment is pledged, realising this collateral in the event of default becomes difficult. Legal proceedings to seize and liquidate collateral are slow and costly, and take a lot of effort, increasing the fixed cost of enforcing small loans (Wenner, 2007; World Bank, 2023). In addition, social and political considerations constrain lenders' willingness to exercise strict enforcement, since large scale foreclosure on poor farmers after a drought can generate reputational damage and community resistance. Given these constraints, formal collateral provides limited effective protection against loss, and expected recovery rates on defaulted rural loans are low. Lenders therefore respond by adopting conservative lending policies, including high interest margins, short maturities, small loan sizes and stringent eligibility criteria, and by rationing credit to borrowers and sectors that are perceived as especially risky (Wenner, 2007; World Bank, 2023). These practices directly limit the availability and affordability of credit for smallholders who wish to invest in new technologies, even when such technologies could improve long term resilience.

Incentive compatibility:

Beyond pure enforcement, lenders attempt to influence borrower behaviour through incentive mechanisms, but these mechanisms are typically simple and are not explicitly tailored to climate risk. Evidence from rural financial inclusion initiatives in Guatemala indicates that lenders commonly reward good repayment performance with access to larger loans, marginally lower interest rates or more flexible repayment schedules, and that they often prefer to work through

group lending structures or cooperatives where peer monitoring encourages timely repayment (FAO, 2018; NCBA CLUSA, 2024). Some institutions condition credit access on participation in technical assistance, agronomic training or financial literacy programmes, which are intended to enhance farm management and reduce the probability of default (NCBA CLUSA, 2024; Center for Financial Inclusion, 2024). However, the strength of these incentive schemes is constrained by lenders' own exposure to climate shocks and by their capital and funding limitations. Because insurance cover is viewed as imperfect and collateral is weak, lenders must retain sizable buffers against potential losses, which restricts their ability to offer significantly more favourable terms to borrowers adopting innovative but risky technologies (Wenner, 2007; World Bank, 2024).

The interaction between portfolio concentration, fragile enforcement and limited incentive instruments generates a structural reluctance among lenders to extend and adapt credit to support technology adoption by smallholders in Guatemala. When drought is the principal driver of default, and when insurance is perceived as subject to high and increasing basis risk, lenders see little assurance that claims will be paid in the very seasons when their borrowers are most likely to default (World Bank, 2024; Carter et al., 2016). In this setting, financing the uptake of new technologies appears to add further risk to already vulnerable portfolios, while the institutional tools available to control that risk remain weak.

4.3 Operational constraints: distribution, servicing, and claims settlement- Max

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- What are the main frictions in client servicing (documentation, literacy, trust, after-sales support)?
- How do current parametric schemes manage enrolment and payouts operationally, and what problems have emerged (e.g. confusion around triggers, delays, network access)?
- For a bundled product, which partner configuration (insurer + MFI + Ministry, etc.) looks most realistic given today's operational capacities?

The following chapter examines how operational constraints in the distribution, servicing and claims settlement of agricultural credit and parametric insurance in Guatemala shape the effective basis risk perceived by smallholders and lenders and thereby limit the role of insurance as a reliable credit enhancement tool.

Distribution channels:

Agricultural credit and insurance for smallholders in Guatemala are delivered through a fragmented set of channels, including public programmes operated by the Ministry of Agriculture and the public development bank, commercial and cooperative banks, specialised microfinance institutions and, in some cases, non-governmental organisations partnering with private insurers (World Bank, 2023; OECD, 2021). These institutions typically rely on sparse branch and agent networks in rural areas, complemented by temporary outreach campaigns at municipal centres and through producer associations (OECD, 2021; Center for Financial Inclusion, 2024). The geographic dispersion of smallholders, poor road infrastructure and security concerns increase the fixed costs of reaching remote communities and limit the frequency of in-person interaction. As a result, reliance on batch registration through local leaders or organisations often introduces errors in client lists and poor alignment between insured units and farm plots. From a basis risk standpoint, these distribution constraints create a mismatch between the insured population and the index design, increasing the likelihood that realised losses diverge from index outcomes.

Client servicing:

Client servicing and communication frictions reinforce these problems. Many smallholders face documentation barriers, including limited formal identification, incomplete land records and low familiarity with formal financial products, while literacy and numeracy levels can be low in some indigenous and rural communities (OECD, 2021; Center for Financial Inclusion, 2024). Front-line

staff in banks and microfinance institutions are often under time pressure and may lack the training needed to explain parametric insurance concepts, such as index triggers, coverage limits, and payouts based on predefined indicators rather than actual losses. In practice, this means that farmers often form expectations about insurance performance that do not match the contractual trigger conditions, particularly regarding when a payout should occur after a drought event. For lenders, incomplete or outdated client data and weak after-sales support result in uncertainty about which borrowers are insured, for what amounts and under which conditions. These service gaps create what is known as "expectation risk." Even if the index accurately reflects average area losses, farmers and lenders may perceive high basis risk because their expectations of coverage are often not met.

Claims settlement:

Operational arrangements for claims settlement in Guatemala's parametric schemes further shape these perceptions. Recent public and donor-supported programmes have used rainfall indices to trigger payouts for smallholder farmers, with enrolment typically handled through lists submitted by municipalities, cooperatives or participating financial institutions. Payouts were delivered via public bank branches, automated teller machines or digital codes sent by text message (World Bank, 2024; WFP, 2021). While the model is designed to enable relatively fast disbursement, multiple recurring issues have been documented in evaluations and practitioner reports. These include delays between the occurrence of a climate shock and the communication of trigger results, mismatches between official beneficiary lists and farmers' current borrowing status, network or access issues that prevent some recipients from receiving or redeeming payout codes, and disputes when farmers experience visible crop damage but the index does not reach the trigger threshold (World Bank, 2024; J-PAL, 2024). Users often cannot tell operational failures apart from downside basis risk. Regardless of whether the cause lies in index design or administrative breakdowns, the outcome is a missed payment when it´s most critical.

Finally, the effective settlement and use of claims within a bundled credit-insurance product depends on coordination between insurers, lenders and, where relevant, public agencies. In Guatemala, early agricultural insurance programmes have often involved a public insurer or intermediary, a development bank and one or more ministries, with donors providing technical assistance and premium support (World Bank, 2024; WFP, 2021). Aligning enrolment cycles, data formats, information systems and reporting requirements across these actors is operationally demanding. In the absence of integrated systems that link loan balances, insurance coverage and trigger outcomes at the client level, lenders may not be able to automatically apply payouts to debt service or to identify which borrowers should receive restructuring offers following an insured event. This weakens the potential of insurance to operate as a reliable credit enhancement tool and leaves both farmers and lenders exposed to residual risk that they interpret as a failure of the insurance mechanism.

4.4 Regulatory Constraints: Strategic role of public policy and safety nets –

The supply-side challenges to the innovative product's implementation in Guatemala are further compounded by the local regulatory environment. The Superintendencia de Bancos (SIB), the country's local insurance regulator, has yet to enact specific laws dedicated to parametric

insurance and currently takes a discretionary, reactive approach to approving index-based insurance scheme. This is primarily because existing national insurance laws were drafted for traditional indemnity models, which require physical damage assessments while index-based products that rely on proxy variables like rainfall thresholds (World Bank, 2024). It's therefore important for future models to consider the impact that formal legislation could have on pricing, scale and predictability of the local insurance and reinsurance market.

Furthermore, the bundling of solutions, such as insurance combined with loans or seeds, introduces questions and concern around the lack of legislation and standardization around datasets or key inputs into models may cause friction with insurer-based estimates, particularly where basis risk materializes and impacts payouts. The model illustrates the severity of this risk, noting that a single "False Negative" event, loss without a payout, can trigger a 55% disadoption rate among farmers. To mitigate this on the supply side, insurers are forced to apply a 15pp farm-scale rho discount to account for idiosyncratic microclimate variations not captured by satellite data and which result. Without regulatory supervision and agreement, however, this technical adjustment may cause frictions between insurers and uptakers.

Finally, although the *Esfuerzo Seguro* program, which bundles index insurance with loans from BANRURAL, and the *Seguro Colectivo Paramétrico* program, where MAGA acts as the policyholder and pays 100% of the premiums on behalf of the farmers, exist, these solutions are unlikely to achieve the scale needed to cope with needs of all local farmers. (World Bank, 2024). Therefore, the government may need to also incentivize private insurers and lenders to provide the bundled insurance scheme through mechanisms such as tax incentives.

4.5 Implications for bundled product design – max

The interaction between insurer, lender and operational constraints therefore sets important boundaries on what a bundled credit–insurance product can realistically achieve in Guatemala and clarifies which frictions subsequent design choices must address. Correlated drought risk, rising basis risk and limited reinsurance capacity restrict insurers to relatively simple, standardised index products, while concentrated loan portfolios, weak collateral and cautious incentive schemes make lenders unwilling to rely on insurance as a robust form of credit enhancement unless payouts are timely, predictable and clearly linked to debt service. At the same time, fragmented distribution channels, servicing gaps and imperfect claims settlement generate additional perceived basis risk for both farmers and lenders, because administrative failures and communication problems often translate into missing payments precisely when they are most needed. These combined constraints motivate a careful examination of evidence from previous index-based insurance experiments in other developing countries, in order to identify which product features and implementation strategies have successfully overcome similar barriers to adoption and effectiveness, and to inform the design choices developed in the following chapter.

5 Evidence from Similar Experiments: Lessons Learned

This section reviews evidence from previous case studies and experiments with comparable interventions using index-based insurance in other developing countries. The effectiveness of bundled index insurance solutions depends critically upon uptake by farmers and the reliability of the insurance mechanism. Evidence from experiments and programmes across developing countries highlight that adoption is not primarily constrained by the absence of insurance products, but other factors such as liquidity constraints, basis risk and behavioural factors may shape both farmer demand and lender participation. The extent of these constraints determine whether index-based insurance can generate meaningful changes in agricultural investment, productivity and climate resilience.

5.1 Mechanisms for increasing uptake

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5.1.1 Liquidity, timing and the 'bundling penalty'

In Ethiopia, the timing of insurance payments was investigated with index insurance bundled with credit and input vouchers in the Rift Valley zone. Studies in Kenya and Uzbekistan examined subsidised and repeated insurance exposure (Bulte et al., 2020; Carter et al., 2025). Delaying payment was found to increase insurance adoption by roughly 19%, whilst bundling delayed premium insurance with credit and input vouchers increased uptake by 32% and significantly boosted investment in fertiliser and improved seeds. Evidence from Kenya and Uzbekistan found that subsidised insurance and repeated exposure increased willingness to invest in high-return inputs and improved long-term participation with bundled insurance.

A field experiment in Malawi offered farmers a choice between standard uninsured loans and loans bundled with weather index-based insurance at actuarially fair rates (Giné & Yang, 2009). The authors found that bundling insurance reduced loan uptake by 13% relative to uninsured loans, highlighting a potential 'bundling penalty' where farmers interpreted the premium on bundled insurance as a higher effective interest rate (Giné & Yang, 2009). This discouraged uptake for smallholders due to constrained liquidity and cash scarcity at the time of planting crops, presenting a trade-off between upfront payments with input purchases (Giné & Yang, 2009).

A key consideration is for our NGO is timing of payment. If premiums are charged upfront, the bundle may suppress rather than increase demand for credit. A viable design might require delayed or post-harvest premiums, financing mechanisms or early-stage subsidies to avoid the 'bundling penalty'.

5.1.2. Unlocking credit supply through meso-level insurance

In Northern Ghana, a randomised a field experiment investigated agricultural loans bundled with index insurance, where payouts were directed either to farmers (micro-level) or to lenders (meso-

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2) model section - inputs and outputs-> informs solution
3) solution/recommendation based on the model, recommending best inputs and why and approach based on examples as well

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level) to cover loan defaults in the event of drought (Mishra et al, 2021; 2023). The experiment found that meso-level insurance significantly increases loan approval rates by 17-25% driven by reduced lender risk exposure.

A study in Uzbekistan examines how index-based insurance functions as a credit-enabling tool for smallholder farmers lacking collateral for financing fertilizer and improved seed adoption (Carter et al., 2025). This study revealed how insurance acted as a substitute for collateral, allowing farmers to access credit and leading to increased investment in high-return inputs. Insured farmers in Uzbekistan were substantially more likely to adopt fertilizer and improved technologies, suggesting that reducing downside risk can increase productive investment. Neither micro-level nor meso-level insured loans significantly increased rates for farmers' loan applications, indicating the presence of persistent demand-side constraints. In Uzbekistan, initial uptake was limited by low understanding and trust in the index-based insurance product which required repeated exposure and experience to build participation. Both case studies highlight how the efficacy of insurance depended on the credibility of payouts. Farmers' uncertainty about contract performance hindered uptake of insured loans.

These findings suggest that the supply-side risk aversion from lenders is likely to be a barrier to expanding input finance, rather than insufficient demand from farmers for credit. A bundled insurance product in Guatemala should therefore incorporate meso-level insurance to protect lenders' portfolios, whilst also ensuring that farmers perceive tangible benefits from the product. Design must account for both demand-side and supply-side, to prevent insurance expanding credit supply without translating to higher uptake and technology adoption for smallholder farmers.

5.1.3. Investment responses and adoption fragility

Beyond initial uptake, a key question is whether insurance promotes sustained investment over time. A multi-year field experiment in Tanzania and Mozambique tested a bundled product that combined drought-tolerant maize seeds with satellite-based index insurance (Boucher et al., 2024). Farmers who experienced a severe shock and received a payout, subsequently expanded their cultivation area by approximately 67% and intensified input use in the following season (Boucher et al., 2024). This demonstrates a 'resilience dividend' with insurance compensating for losses whilst also accelerating investment. Farmers who purchased the bundled insurance but did not experience a shock and therefore did not receive payout frequently abandoned the technology in subsequent periods, showing that adoption was highly fragile (Boucher et al., 2024).

The fragility of adoption in Tanzania and Mozambique highlights an area of concern that should be considered for Guatemala (Boucher et al., 2024). During the initial years of having the bundled insurance, farmers who pay for the product but experience no drought and therefore receive no payout may interpret the premium for the bundle as a loss or unnecessary cost, especially if their neighbours who may not have paid the premium are better off. Factors such as risk aversion and present bias may amplify the risk of this in Guatemala for smallholder farmers. Our NGO could

provide sustained, early-stage subsidies for an ‘experiential’ phase in the first two to three seasons to gain farmers’ trust in the product. Additionally, investment in educating farmers’ on the benefits of insurance as a protection measure against catastrophic climate events.

5.2 Addressing basis risk and trust

Basis risk and the pervasive lack of farmer trust represent the primary barriers to the successful adoption of bundled index insurance and input finance products (Biffis *et al.*, 2022). If a smallholder farmer assumes credit to invest in expensive, climate-resilient technologies but suffers a localised shock that the index fails to capture, known as downside basis risk, they face the compounded devastation of loan default, lost premiums, and complete crop failure. This scenario destroys demand-side trust and exacerbates supply-side portfolio risk for lenders. The following subsections detail how international case studies navigate these challenges, offering direct implications for the NGO ´s strategy in Guatemala.

5.2.1 Mitigating Spatial Mismatch and Securing Trust

In Kenya's Kilimo Salama programme, automated solar-powered weather stations were deployed to collect highly localised, objective rainfall data, practically eliminating geographical mismatch and fostering trust among previously sceptical farmers (International Finance Corporation, 2010). Experimental evidence from Kenya quantitatively supports this mechanism by showing that in the case that farmers trust the objective measurement of the insurance product, unconditional input investments increase significantly (Bulte *et al.*, 2020). Specifically, offering free multi-peril insurance bundled with certified seeds increased farmers' total non-seed investments, such as fertiliser and hired labour, by almost 1,700 Kenyan Shillings, representing a statistically significant increase of 0.15 standard deviations (Bulte *et al.*, 2020).

While Kenya successfully utilised highly localised ground-based weather data, Guatemala's current ground weather station network is far too sparse to support reliable weather index insurance, forcing domestic programmes to rely heavily on satellite imagery (World Bank Group, 2024). Currently, Guatemala's Ministry of Agriculture, Livestock and Food (MAGA) implements the Seguro Colectivo Paramétrico, which relies on ERA5 satellite data operating on a coarse 27x27 kilometre resolution grid (World Bank Group, 2024). This coarse resolution fails to capture Guatemala's highly complex topography and microclimates, leading to significant spatial basis risk and potential mistrust. Since deploying a dense network of physical stations across mountainous terrain is difficult, the NGO must approximate Kenya's localised accuracy by transitioning from ERA5 to high-resolution satellite datasets like CHIRPS (World Bank Group, 2024). Operating at an approximately 5.5x5.5 kilometre resolution, CHIRPS has been successfully applied to characterise agricultural drought conditions in the region and can provide the objective, granular data needed to foster farmer trust on the demand side and reassure lenders on the supply side (World Bank Group, 2024).

5.2.2 The Impact of Local Basis Risk on Technology Adoption

Beyond spatial mismatch, farmers are acutely sensitive to local basis risk, which occurs when a policy explicitly excludes non-weather perils such as pest infestations (Freudenreich & Mußhoff, 2018). A framed field experiment with Mexican maize farmers revealed that index insurance plagued by local basis risk performed noticeably worse than full indemnity insurance in driving

the adoption of hybrid seeds (Freudenreich & Mußhoff, 2018). Furthermore, a 50% premium subsidy completely failed to increase adoption under the index insurance scheme, whereas it succeeded under the indemnity scheme (Freudenreich & Mußhoff, 2018).

Even with an accurate index, poor cost-allocation can actively suppress adoption. A randomised experiment in Malawi found that bundling a hybrid seed loan with weather index insurance reduced loan take-up by 13 percentage points, dropping from a 33.0% baseline (Giné & Yang, 2009). Because these highly impoverished farmers already had "implicit insurance" through limited liability, meaning they faced few consequences for defaulting on a standard loan during a drought, forcing them to purchase formal insurance was perceived purely as an increased interest rate rather than a protective benefit (Giné & Yang, 2009).

Guatemalan family farmers face multifaceted biological threats beyond drought, such as coffee rust and pests (World Bank Group, 2024). Furthermore, the Malawi case serves as a crucial warning against poor product design. Simply adding the insurance premium as a mandatory upfront cost will likely suppress credit demand. Because smallholders in Guatemala often operate on communally held land without formal titles, lenders already struggle to seize collateral, giving farmers a degree of limited liability. To avoid pricing farmers out, the NGO must not treat insurance merely as an add-on cost. Instead, the bundle should feature heavily subsidised or delayed premiums, or utilise "meso-level" insurance that protects the lender's portfolio directly, allowing the farmer to benefit from lower interest rates without paying a separate premium. Simultaneously, to cover the local basis risk gaps, the index insurance must be closely bundled with agronomic training or pest-resistant seed varieties.

5.2.3 Enhancing Trust Through Audits and Premium Structuring

Managing the perception of basis risk and overcoming severe liquidity constraints are essential for sustaining trust over time. In Ethiopia, a randomised control trial demonstrated that allowing farmers to delay premium payments until after the harvest and bundling index insurance with credit and an input voucher surged baseline uptake by 32 percentage points (Belissa, Lensink & Marr, 2025). To maintain trust and prevent defaults under this delayed premium scheme, the Ethiopian programme instituted an independent agronomic audit to transparently verify losses whenever farmers complained the index had failed them (Belissa, Lensink & Marr, 2025). This transparent process kept defaults remarkably low, with only 29 out of 459 purchasing households, approximately 6%, defaulting (Belissa, Lensink & Marr, 2025). This is supported by findings of a multi-year trial in Tanzania and Mozambique, where farmers who received an insurance payout after a severe yield shock exhibited a "resilience dividend," significantly increasing their input expenditures by $129.50 and expanding their maize area by 1.5 hectares in the following season (Boucher *et al.*, 2024).

To effectively manage basis risk disputes and ensure fairness when the index fails to capture localized shocks, the Guatemalan programme should incorporate conditional, ground-level agronomic audits directly into the contract (World Bank Group, 2024). By combining this transparent fail-safe audit with a delayed premium structure and input vouchers, the NGO can build the necessary demand-side trust to drive high adoption rates and secure long-term agricultural resilience.

6 Policy Implications and Design Recommendations

6.1 Feasibility assessment

The model gives a clear answer on feasibility: the bundled product is not viable at unsubsidised market rates, but it becomes financially meaningful for farmers under a fifty percent premium subsidy, combined with a CHIRPS-quality index and conditional audits. The next paragraphs set out the conditions that make this possible and the constraints that still bind.

Under fair actuarial pricing with a thirty-five percent commercial loading factor, every scenario without subsidy produces negative income for the farmer. The ERA5 baseline (S1) gives a net loss of nineteen dollars per hectare per year; CHIRPS without subsidy (S2) still leaves a loss of thirteen dollars. This is not a modelling quirk. With a commercially viable loss ratio of 0.74×, the premium of roughly eighty-four dollars per hectare under CHIRPS exceeds the expected payout value after loan interest. That is consistent with J-PAL (2024) finding that uptake at market prices rarely reaches twenty percent, and it confirms that subsidy is a structural prerequisite, not an optional add-on.

A fifty percent premium subsidy under CHIRPS with conditional audits (S10) changes the result: income rises by twenty-nine dollars per hectare per year, while per-season default falls to 1.0 percent and lifetime default over twenty seasons to 20.7 percent. The insurer still stays commercially viable at a loss ratio of 0.74×, and the NGO's annual cost is fifty-five dollars per farmer, made up of a forty-two dollar subsidy and thirteen dollars in audit costs. For forty thousand insured smallholders, that implies about 2.2 million dollars per year, which sits within the range of multilateral agricultural finance programmes.

The model points to five enabling conditions. The index must reach at least $\rho \geq 0.65$ at farm scale (Biffis and Chavez, 2017): ERA5 ($\rho_{tam} = 0.30$) falls well short, CHIRPS ($\rho_{tam} = 0.63$) comes close but does not quite meet it, and the NDVI-SPI composite ($\rho_{tam} = 0.70$) clears it. A formal lender, BANRURAL or a comparable rural cooperative, must take part for the Carter et al. (2016) portfolio effect to work. MAGA extension services must keep distributing drought-tolerant seed for at least three seasons to allow the technology adoption trajectory to develop. Multi-year NGO funding of at least five seasons must be committed in advance, because a programme that withdraws subsidy after one or two seasons will generate a trust collapse rather than a graduation toward self-financing. Finally, the regulatory framework must accommodate interlinked insurance-credit contracts, which Guatemala's current insurance legislation, drafted for traditional indemnity products, does not yet explicitly permit.

Three constraints are binding. Guatemala's station network, with 97 stations over 109,000 km^2, is too sparse for meso-level calibration below departmental scale. The model's $\rho = 0.78$ for

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CHIRPS is also borrowed from Mozambique, because no Guatemala-specific estimate exists; if the true Dry Corridor value is closer to 0.60 or lower after the farm-scale discount, feasibility weakens sharply. Finally, the climate non-stationarity stress test (S9, pd = 50%) produces a lifetime default probability of 100 percent and a VaR of minus eighty-seven dollars per hectare, showing that the current design would fail under a large shift in drought frequency. Regular recalibration is therefore essential.

6.2 Product design choices/recommendations

The ten scenarios cover the main design dimensions at once, so the recommendations below rest on quantitative comparison rather than general principle.

Index choice and minimum viable correlation threshold

The key decision is which rainfall index to use. ERA5 at 27×27 km gives a farm-scale ρ of 0.30, far below the Biffis and Chavez (2017) threshold of 0.65, a false-negative rate of 14.5 percent, and a five-percent Value-at-Risk of twelve dollars per hectare. CHIRPS at 5×5 km improves ρ to 0.63, lowers false negatives to 10.4 percent, and raises VaR to thirty-one dollars. The NDVI-SPI composite reaches ρ_{tam} = 0.70, clears the minimum threshold, and gives a false-negative rate of 9.6 percent and VaR of thirty-four dollars. The NGO should treat CHIRPS as the minimum acceptable index and move to the composite within three seasons once NDVI integration is in place. Staying with ERA5 is not conservative; it is a decision to deliver a product that will fail in roughly fourteen out of every one hundred drought years.

Premium timing and harvest-linked deduction structure

The Malawi experiment (Giné and Yang, 2009) showed that collecting the premium at planting, when liquidity is tightest, reduced uptake by thirteen points. Evidence from Ethiopia shows that delayed or harvest-linked collection increased adoption by about nineteen percent and raised fertiliser and seed investment by thirty-two percent. The practical implication is simple: the premium should be deducted from harvest-season loan repayment, not charged at planting. Under S10, the farmer pays forty-two dollars per hectare, the fifty-percent co-payment on the fair premium of eighty-four dollars, at harvest rather than at loan origination. That avoids the bundling penalty seen in Malawi and matches the payment to the season when the farmer has either received revenue or received a payout.

Meso-level interlinkage versus standalone product structure

Carter, Cheng and Sarris (2016) show that insurance linked to the lender's portfolio creates a 1.39× income multiplier relative to standalone insurance by reducing the risk premium in the lending rate. In the model, that appears as a credit-rate cut from twenty-two to 17.3 percent under CHIRPS (S2, S10), which already includes an eight-dollar per hectare interest saving (Biffis et al., 2022). The mechanism only works if drought payouts are routed at least partly to loan repayment rather than paid entirely as cash. A pure lender-first structure gives no visible benefit to farmers, while a purely cash-based payout weakens the lender's incentive to participate.

Subsidy design: the S7 versus S10 trade-off

This is the most consequential design choice the NGO must make, and the model presents it as a policy judgement rather than a technical optimum. On every quantitative dimension, full subsidy (S7) strictly dominates fifty-percent subsidy (S10): income gain of seventy-one versus twenty-nine dollars per hectare, per-season default of 0.3 versus 1.0 percent, twenty-season lifetime default of 5.3 versus 20.7 percent, and VaR of 116 versus 73 dollars. Year-five adoption is essentially identical at 42.5 versus 42.4 percent, confirming that the adoption ceiling is determined by trust and seed availability, not by the residual premium level within the subsidised range. The quantitative case for S7 is unambiguous. The sustainability case for S10 rests on considerations the model does not capture. A zero co-payment (S7) creates no contractual relationship between farmer and insurer, generates no renewal habit, and makes the programme entirely dependent on uninterrupted NGO funding with no pathway to graduation. The forty-two dollar annual co-payment in S10 establishes a commercial relationship, creates a demonstrated willingness-to-pay signal for future market development, and allows the NGO to begin a subsidy step-down trajectory in Years 3 to 5 without catastrophic demand collapse. The honest position is that S7 is the superior welfare programme and S10 is the superior development finance programme. The NGO must state this trade-off explicitly in its donor reporting.

Basis risk management: conditional audits versus Basis Risk Fund

The model compares S4 (ERA5 plus BRF) and S5 (CHIRPS plus conditional audits). Without subsidy, both produce negative income gains, but the BRF adds two dollars per year to NGO costs without changing adoption, while audits add thirteen dollars per year and lift year-five adoption from 23.1 to 36.4 percent. The audit premium comes from the trust-recovery effect in Mobarak and Rosenzweig (2013): each objective audit strengthens confidence across the peer network. The BRF, by contrast, pays individuals but does not build community trust and could be depleted quickly in a severe El Niño. The conditional audit protocol is therefore the better basis-risk tool.

Addressing peril basis risk: the named-peril rider

The model's false-negative rate captures only weather-index mismatch, not losses from fall armyworm, coffee leaf rust, local flooding or frost. De Janvry et al. (2014) show that farmers tolerate geographical basis risk more readily than local uninsured losses, and that excluding non-weather perils depresses adoption even when the insurance is otherwise attractive. The NGO should therefore include a named-peril rider for fall armyworm and Roya from Year 1 in pilot areas, with the rider fully subsidised in Years 1 and 2 to remove early friction.

Trust-building mechanisms: the guaranteed payout and peer network

Mobarak and Rosenzweig (2013) show the problem clearly: farmers in strong informal networks see no obvious benefit in non-drought years and may view the premium as a loss. Three steps can break that pattern. First, a guaranteed first-season payout gives the product an immediate visible payoff. Casaburi and Willis (2018) show this works better per dollar than subsidy alone in low-trust settings. Second, a public payout register, at municipal and cooperative offices and online, makes each trigger and payout easy to verify. Third, two or three respected farmer champions per village can spread trust more efficiently than a broader marketing campaign.

Drought-tolerant seed complementarity: why bundling is not optional

Comparing S2, where CHIRPS is bundled with drought-tolerant seed and no subsidy, with S8, where CHIRPS is bundled with conventional seed and no subsidy, can make it look as though drought-tolerant seed lowers welfare. That is the wrong reading. The larger loss in S2 reflects the higher fair premium on a larger insured input package, not a failure of the seed itself. Once subsidy is added, S7 delivers a seventy-one dollar per hectare gain because drought-tolerant varieties raise yields in both severe and normal drought years. Without them, the insurance protects a weaker crop and the resilience dividend described by Biffis et al. (2022) does not emerge. Drought-tolerant seed should therefore be mandatory in the input loan, not an optional extra.

6.3 Implementation Roadmap

The roadmap follows the natural programme sequence of validation, trust-building and institutional graduation. The phase boundaries are indicative, but the escalation triggers in Section 6.5 define when the programme can move on.

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Years 1–2 (Pilot and Index Validation)

The first two seasons are for validation and trust-building, not scaling. The NGO should work with 500 to 1,000 farmers in three to four municipios in Chiquimula and Jalapa, where drought exposure is strongest and which overlap with the WFP-Aseguradora Rural pilot footprint. CHIRPS should operate as the main payout trigger, with ERA5 recorded in parallel for comparison. NDVI should be fed in alongside agronomic yield monitoring at twenty sample farms per municipio to build the ground-truth dataset needed for composite calibration. Subsidy should be eighty percent in Years 1 and 2, above the fifty-percent S10 level, so the first cohort is very likely to see a positive outcome. The named-peril rider for fall armyworm and Roya should be fully subsidised. Baseline data should be collected at enrolment on income, adoption, weather readings and risk preferences.

Years 2–4 (Scaling and Subsidy Step-Down)

If the pilot meets the index-quality threshold (validated $\rho \geq 0.65$), delivers at least one visible payout, and retains more than seventy percent of farmers in Year 2, the program can expand to Jutiapa, Zacapa and El Progreso. Subsidy then steps down from eighty percent to fifty percent at the start of Year 3. BANRURAL enters as the lending partner under a tripartite agreement with the insurer and the NGO, activating the Carter et al. (2016) portfolio effect. Conditional audits become formalised, with a maximum thirty-day response time. A second lender, ideally a rural cooperative federation, should join in Year 3 to keep pressure on lending rates. NDVI-SPI can become the main trigger once the pilot calibration dataset is large enough to validate it empirically.

Year 4 and beyond (Institutional Sustainability)

From Year 4 onward, the program should begin a planned step-down toward a long-run subsidy target of twenty-five to thirty percent, financed through government agricultural support, climate-adaptation finance and a levy tied to falling insurer losses as drought-tolerant seed adoption rises. An Index Calibration Committee, INSIVUMEH, MAGA, the insurer's actuary and

an independent academic, should review triggers each year. An independent impact evaluation should report in Year 4 so donors can fund later tranches on performance rather than process. The program is not expected to become fully self-financing within five years without public support, but the goal is to cut NGO cost per farmer from fifty-five dollars in Year 3 to below thirty dollars by Year 6.

6.4 Conditions for success

The following seven conditions are necessary: if ~~any one~~anyone fails, the program becomes either financially unviable or institutionally weak. Each condition is grounded in the model or the evidence reviewed in Section 5.

1. Index quality must reach ρ ≥ 0.65 at farm scale. Biffis and Chavez (2017) set this as the minimum for positive welfare effects among risk-averse farmers. ERA5 (ρ_{tam} = 0.30) fails outright and should be replaced by CHIRPS (ρ_{tam} = 0.63) as a transition index and the NDVI-SPI composite (ρ_{tam} = 0.70) as the target standard.
2. Formal lender participation must be real, not assumed. The Carter et al. (2016) multiplier only works if a lender actually lowers rates as portfolio risk falls. The NGO should check this each year; if lenders keep the benefit as margin instead of passing it through, the twenty-nine dollar per hectare gain under S10 is overstated and the feasibility case weakens.
3. MAGA extension services must remain continuous. Drought-tolerant seed distribution depends on the MAGA network, which has been interrupted before by fiscal tightening. The adoption model assumes uninterrupted supply; if extension stops in any season, adoption will fall below the 42.4 percent Year-5 projection and the actuarial gain will shrink.
4. A multi-year funding commitment of at least five seasons must be secured before launch. Boucher et al. (2024) show that adoption is fragile. If subsidy ends too early, farmers are left with a bad first impression before trust has time to build. Funding should therefore be structured as a multi-tranche facility with performance triggers, not annual renewals.
5. At least one visible trigger payout must occur within the first two seasons. The model's trust-recovery parameter only compounds if farmers see a payout. Mobarak and Rosenzweig (2013) show that non-drought seasons are where disadoption starts. If Seasons 1 and 2 are both dry-free, the NGO should consider a guaranteed first-season transfer as part of the insurance cost.
6. Regulatory approval for the interlinked contract structure must be in place before launch. Guatemala's current insurance law, written for traditional indemnity products, does not clearly allow parametric triggers or routing payouts to loan repayment. The NGO should work with the Superintendencia de Bancos in Year 1 to secure a sandbox-style exemption, as Mexico did for CADENA.
7. A climate recalibration protocol must be in place before the first El Niño, not after it. The high-stress scenario (S9, p_e = 50%) shows that a sharp shift in drought frequency drives lifetime default to one hundred percent and VaR to minus eighty-seven dollars. Trigger thresholds based on the current thirty percent drought probability will under-trigger if

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conditions shift, so the Index Calibration Committee must meet every year and escalate automatically if drought frequency exceeds 40 percent over any three-season window.

6.5 Monitoring, evaluation, and learning

The MEL framework rests on five KPIs, an annual recalibration schedule, an audit trigger protocol and a two-tier default monitoring system that keeps actuarial and programme-assessment metrics separate.

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6.5.1 Key Performance Indicators

The five primary indicators and their numerical targets are as follows.

- False-Negative Rate: target ≤ 10% by the end of Year 2 and ≤ 8% by the end of Year 4. CHIRPS is the baseline at 10.4%, and the composite target is 9.6%. Any season above 14.5%, the ERA5 level, should trigger an immediate review. World Bank DRIFCA (2024) recommends treating basis-risk incidents as an early reputational indicator, not just a financial one.
- Year-5 DT Seed Adoption Rate: target ≥ 35% (the model's central estimate under S10 is 42.4%; the conservative lower bound is about 30%). Adoption should be split by gender and indigenous status. WFP Guatemala and IFAD INSURED (2022) report that women make up about eighty percent of insured farmers in current programmes and nearly thirty percent are indigenous, so those groups need close tracking.
- Insurer Loss Ratio: target 0.70–0.80×, with 0.74× as the S10 value. A ratio above 1.0× means the index is over-triggering and becoming unsustainable; below 0.60× suggests under-triggering and reputational risk from uncompensated farmers. Annual loss-ratio review should be the main actuarial control.
- Per-Season Default Probability: target ≤ 1.0% (S10), with 3.0% as the red-flag threshold (the S1 level). This is the actuarial metric used for pricing review and should be tracked quarterly with lender portfolio-at-risk data, then reconciled with the model annually.
- Five-Year Horizon Default Probability: target ≤ 25% over twenty seasons, with S10 at 20.7%. This is the program-assessment metric used for donor reporting. It is calculated as $P(default\ ever) = 1 - (1 - p_s)^T$ and should be reported each year using observed lender data as the program matures.

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6.5.2 Index Recalibration Schedule

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The Index Calibration Committee should meet every October after harvest to compare CHIRPS rainfall with agronomic yield data from the twenty sample farms in each municipio. If estimated ρ falls below 0.60 in any department for two straight seasons, the committee must adjust the trigger within thirty days or pause new enrolment there. If drought frequency exceeds 40 percent over any three-season window, p_e must be updated immediately and the fair premium recalculated.

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6.5.3 Audit Trigger Protocol

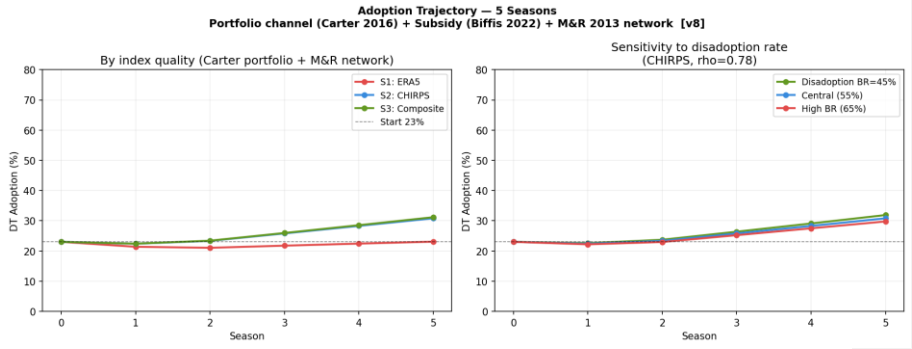
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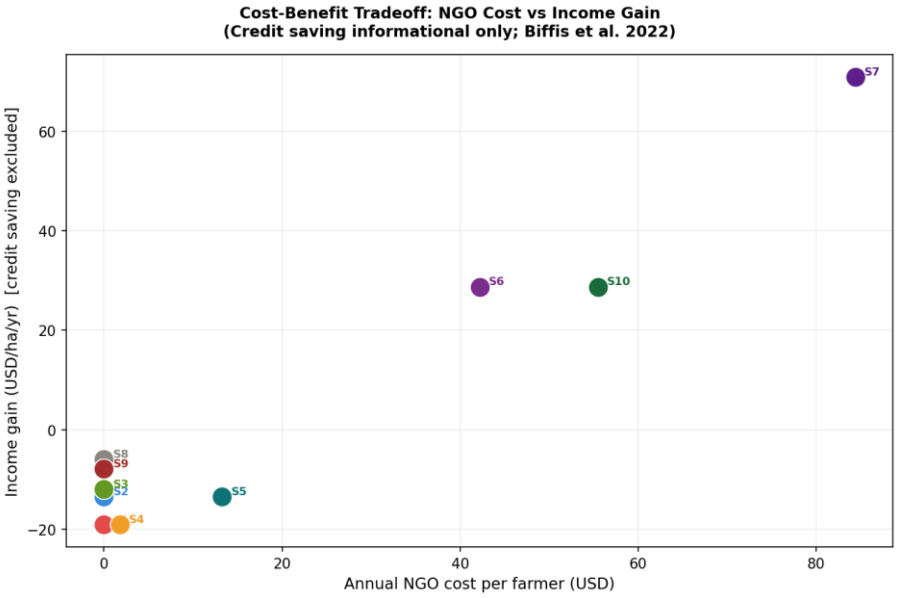
A conditional agronomic audit is triggered when (a) CHIRPS does not breach the trigger in a season and (b) more than fifteen percent of surveyed farmers in a municipio report yield losses above twenty-five percent of expected harvest. If both conditions hold, MAGA sends an independent agronomic team within thirty days to compare ground yields with satellite vegetation data. Findings are reported publicly within forty-five days and posted to the payout register. If the audit shows basis risk affected more than twenty percent of insured farmers, the Index Calibration Committee meets in emergency session and the NGO can use a reserved trust fund, capped at ten percent of annual premium revenue, for a discretionary supplemental payment. The protocol closely follows the Ethiopia programme, where transparency cut the threatened defaulting group to 29 out of 459 farmers (6.3 percent).

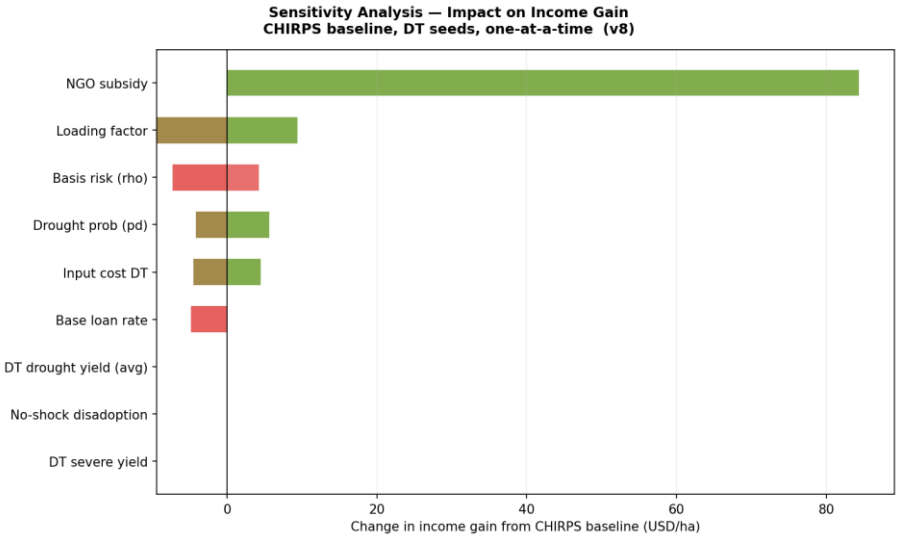
6.5.4 Two-Tier Default Probability Monitoring Framework

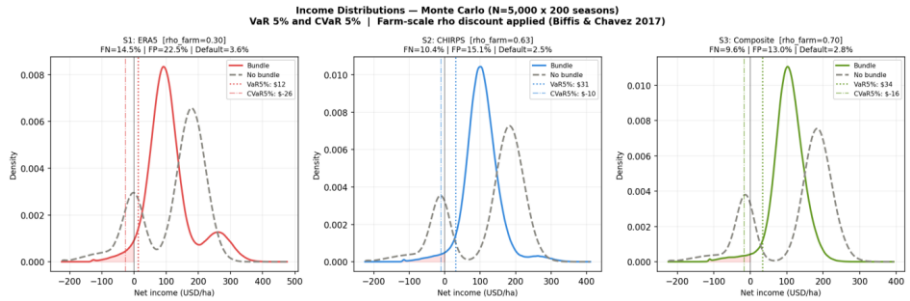
The model gives two different default metrics, and they should not be mixed up in reporting. Per-season default probability is the actuarial metric for pricing review. Under S10 it is 1.0 percent, and the insurer should compare it with the 3.0 percent adjustment trigger. Lifetime default probability over a twenty-season horizon is the donor-facing metric for household welfare. Under S10 it is 20.7 percent, meaning about one in five enrolled farmers will face at least one negative-income season over a typical five-year programme window. That figure should be reported honestly to donors; it is not failure, but a reminder that the product only partially covers weather risk. The quasi-experimental evaluation design, comparing credit-only, insurance-only and bundled cohorts against the existing MAGA Seguro Colectivo Paramétrico as a natural comparison group, following the Mishra et al. (2023) Ghana protocol, provides the counterfactual needed to assess whether 20.7 percent is an improvement over the uninsured baseline.

Taken together, the five KPIs, annual recalibration, conditional audits and two-tier default monitoring system create a learning loop designed to catch failure early, separate actuarial problems from design problems, and build the evidence base needed for decisions on continuation, subsidy step-down and geographic expansion.

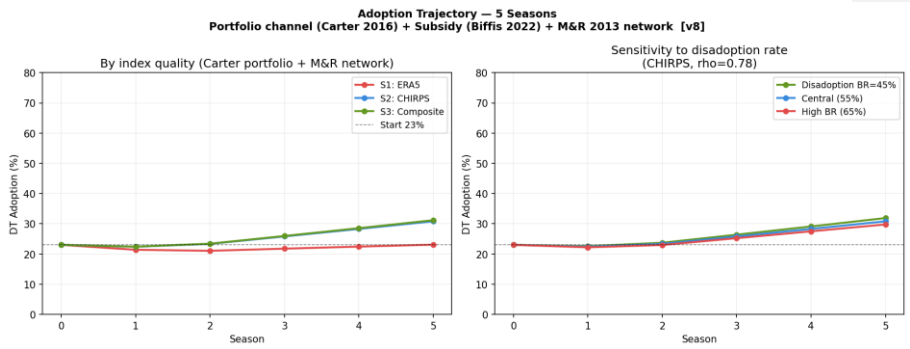


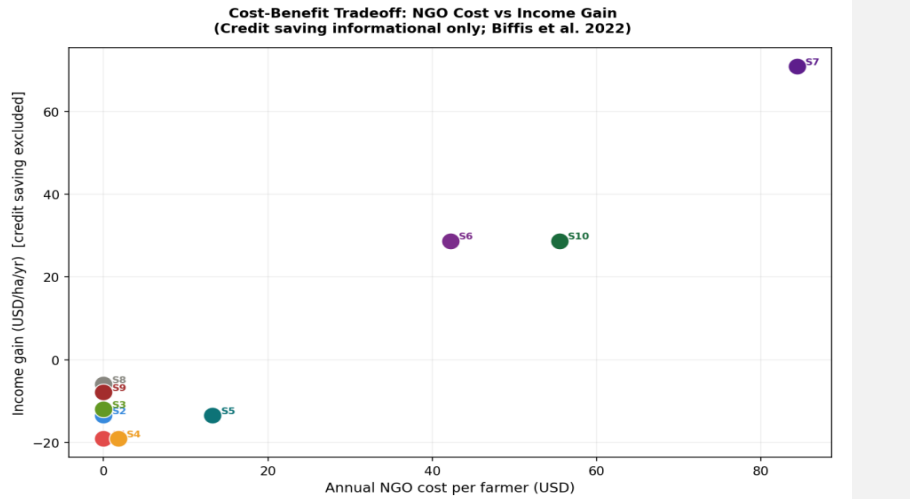

Cost-Benefit Tradeoff: NGO Cost vs Income Gain
(Credit saving informational only; Biffis et al. 2022)
Income gain (USD/ha/yr) [credit saving excluded]
Annual NGO cost per farmer (USD)
S7
S6
S10
S5
S8
S9
S3
S2
S4
60
40
20
0
−20
0
20
40
60
80

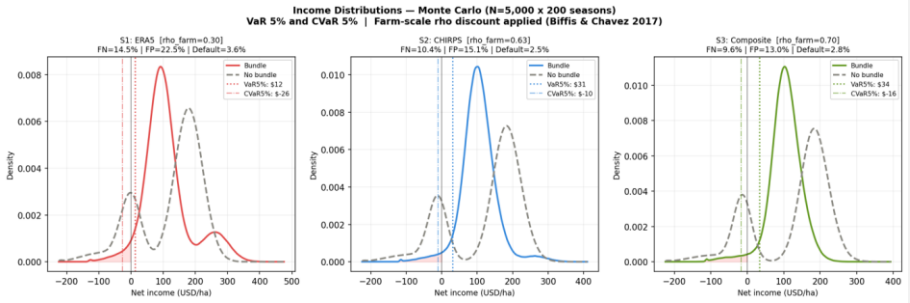

Sensitivity Analysis — Impact on Income Gain
CHIRPS baseline, DT seeds, one-at-a-time (v8)
NGO subsidy
Loading factor
Basis risk (rho)
Drought prob (pd)
Input cost DT
Base loan rate
DT drought yield (avg)
No-shock disadoption
DT severe yield
0
20
40
60
80
Change in income gain from CHIRPS baseline (USD/ha)

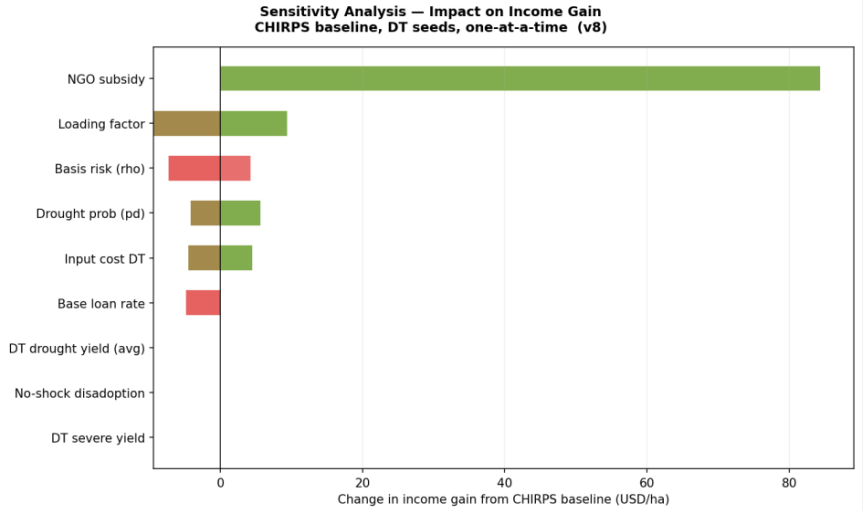

Income Distributions — Monte Carlo (N=5,000 x 200 seasons)
VaR 5% and CVaR 5% | Farm-scale rho discount applied (Biffis & Chavez 2017)
S1: ERA5 [rho_farm=0.30]
FN=14.5% | FP=22.5% | Default=3.6%
S2: CHIRPS [rho_farm=0.63]
FN=10.4% | FP=15.1% | Default=2.5%
S3: Composite [rho_farm=0.70]
FN=9.6% | FP=13.0% | Default=2.8%
Bundle
No bundle
Density
Net income (USD/ha)

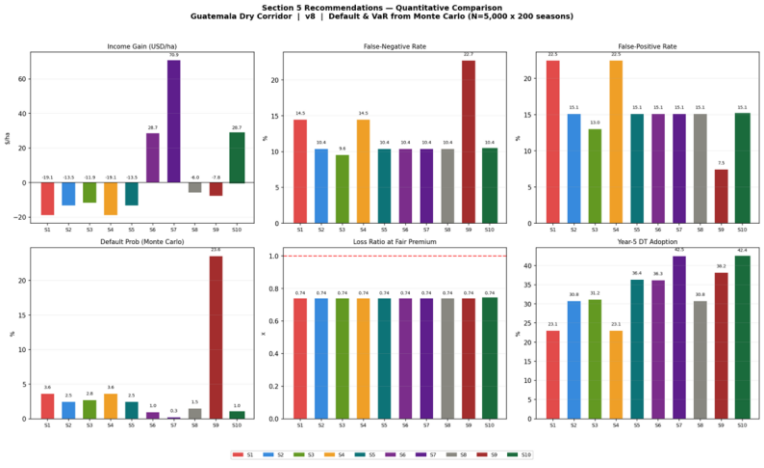
Conclusion


Adoption Trajectory — 5 Seasons
Portfolio channel (Carter 2016) + Subsidy (Biffis 2022) + M&R 2013 network [v8]
By index quality (Carter portfolio + M&R network)
S1: ERA5
S2: CHIRPS
S3: Composite
Start 23%
Sensitivity to disadoption rate
(CHIRPS, rho=0.78)
Disadoption BR=45%
Central (55%)
High BR (65%)
DT Adoption (%)
Season


Cost-Benefit Tradeoff: NGO Cost vs Income Gain
(Credit saving informational only; Biffis et al. 2022)
Income gain (USD/ha/yr) [credit saving excluded]
Annual NGO cost per farmer (USD)
S7
S6
S10
S5
S9
S3
S2
S4







What We Could Have Done Further

The extensions we considered but didn't implement to show we understand the model's boundaries etc

1. Dynamic Premium Repricing

Right now the premium is computed once and stays fixed. But as more farmers adopt DT seeds, their yield distribution changes, they lose less in droughts, so the insurer pays out less, and the fair premium should drop over time. We don't capture this feedback loop. A more complete version would reprice each season based on the share of DT vs conventional farmers in the portfolio, creating a virtuous cycle where adoption lowers premiums which further boosts adoption. Biffis et al. (2022) flag exactly this, they warn that if insurers fail to update their models as seed varieties change, they'll systematically misprice risk.

2. Spatial Heterogeneity Across Sub-Regions

We treat the Dry Corridor as one zone with a single drought probability (30%) and a single rho. In reality it spans multiple microclimates and Guatemala has only 97 weather stations for 109,000 km2. Splitting the population into 2-3 agro-ecological zones with distinct parameters would show where the product works best, capture the diversification benefit for the insurer, and make the basis risk analysis more realistic since a 0.78 rho for CHIRPS is an average that hides a lot of spatial variation.

3. Meso-Level Insurance (Payouts to Lender)

Our model only has micro-level insurance paid to the farmer. But the Ghana experiment (Mishra et al. 2021, discussed in our Section 5.1.2) found that directing payouts to the lender instead increased loan approval rates by 17-25%. Including this would mean modelling the lender's portfolio loss distribution explicitly and showing how meso insurance compresses the tail of correlated defaults. It would make the credit enhancement channel much more convincing than the simple rate formula we currently use.

4. Non-Weather Perils (Pests, Disease)

We capture basis risk from weather-index spatial mismatch (the rho discount) but completely ignore local perils like pests or disease. The Mexico experiment (Section 5.2) found this "local basis risk" was the primary barrier to trust, more than geographical mismatch. Adding a stochastic pest shock that causes yield loss without triggering the weather index would make the FN rate endogenous to the full peril mix and strengthen the argument for bundling insurance with complementary strategies like pest-resistant seeds.

5. Heterogeneous Farmers

All 5,000 simulated farmers are identical: same wealth, same risk aversion, same plot size. Introducing agents with different CRRA coefficients (the Mexico experiment estimates mean CRRA of 1.61 with significant variation) and initial wealth levels would let us identify which segments benefit most and which are excluded even after subsidy. The NGO needs to know whether the product reaches the poorest 20% or only the upper tier of smallholders.

6. Bayesian Trust Updating

Our adoption model uses fixed disadoption parameters calibrated from the literature. A more rigorous approach would model farmer beliefs as a Bayesian posterior that updates after each season's outcome: two correct payouts in a row push trust up, a false negative pushes it sharply down. This would naturally produce the adoption fragility Boucher et al. (2024) documented without us having to hand-set coefficients. It's essentially a full agent-based model though, which would have required a different architecture entirely.

7. Multi-Crop Extension

The report covers maize/bean, coffee, and sugarcane farmers, but the model only does maize/bean. Coffee would bring frost risk and 3-4 year maturation cycles, sugarcane would bring hurricane risk with a very different loss distribution (low frequency, catastrophic severity). A multi-crop model would show the diversification benefit for the insurer across hazard types and inform the NGO's geographic rollout sequencing. Realistically though, this is a separate paper.

8. Bootstrap Confidence Intervals

We report point estimates for VaR, CVaR, and default probability with a seed-stability check showing acceptable spread. But we don't report formal confidence intervals. A bootstrap resampling of the 5,000 x 200 income matrix with 90% bands around each KPI would make the uncertainty in tail estimates explicit. This matters because the policy recommendation rests on specific numbers like "default prob = 1.0%" and the reader should know how precise those are. Standard technique, we just ran out of time.

Figure Explanations

Figure 1 - Scenario Dashboard (6 panels)

This is basically the main figure of the model. It compares all 10 scenarios side by side on 6 key metrics. The idea is that you can see at a glance which design works and which doesn't. Income gain (top-left) shows that without subsidy, the bundle actually costs the farmer more than it helps, S1 through S5 are all negative. It's only once you add a 50% subsidy (S6, S10) that the gain turns positive. That's not a bug, it's exactly what Biffis et al. (2022) found in Tanzania: the actuarially fair premium eats up the benefit unless the NGO steps in. FN and FP rates (top-middle and top-right) capture basis risk, which is the central challenge the brief asks about. ERA5 has the worst FN (14.5%) because its 27km grid misses localized droughts, CHIRPS brings it down to 10.4%. Default prob (bottom-left) is the lender's key metric, the supply-side concern. Loss ratio (bottom-middle) proves the insurer can survive: all bars are at 0.74x, safely below the 1.0x break-even line. In v7 this was 4-9x which made zero sense. Adoption (bottom-right) shows S10 reaching 42% from a 23% baseline, proving the product gains traction over time.

Figure 2 - Income Distributions (Monte Carlo)

This one shows the full income distribution for ERA5, CHIRPS, and Composite. The solid line is with the bundle, the dashed grey line is without. The red shaded area below zero is where the farmer defaults. ERA5 has this weird bimodal shape (two humps) because its 22.5% false-positive rate creates random windfall payouts that distort the distribution. It looks like more upside but it's actually insurer-funded noise, not real welfare. CHIRPS and Composite have much cleaner, tighter shapes. The VaR 5% and CVaR 5% lines mark the worst-case tail: CHIRPS ($31 / -$10) beats ERA5 ($12 / -$26) on downside protection. This figure makes risk visible for both the farmer (how bad can it get?) and the insurer (how spread out are my payouts?), which covers both sides of the brief.

Figure 3 - Sensitivity Tornado

Classic one-at-a-time sensitivity. We take the CHIRPS baseline, vary each parameter from low to high, and measure the impact on income gain. NGO subsidy dominates everything with an $85/ha swing, which tells the reader that the single most important lever is how much the NGO subsidizes the premium, not some obscure actuarial parameter. Loading factor and rho come next. What's also telling is that yield parameters (DT drought yield, severe yield) barely register, meaning the bundle's viability is fundamentally a financial design problem, not an agricultural one. That's a strong insight for a risk management course.

Figure 4 - Cost-Benefit Tradeoff

Scatter plot with NGO cost on x and income gain on y. Each dot is a scenario. The story is instant: S1-S5 cluster in the bottom-left (cheap but negative gain, doesn't work). S6 and S10 sit in the sweet spot (moderate cost, positive gain). S7 is top-right (best farmer outcome but most expensive). S10 costs about $13/yr more than S6 because of audits, but buys you 6 extra percentage points of adoption (42% vs 36%, visible in Figure 1). That extra spend is worth it because early-stage adoption is fragile (Boucher et al. 2024). This is basically the NGO's decision tool.

Figure 5 - Adoption Trajectories

Left panel shows adoption over 5 seasons by index quality. CHIRPS and Composite grow from 23% to ~31%, ERA5 flatlines because its high FN rate keeps destroying trust (Mobarak & Rosenzweig 2013 peer effect: farmers who don't get paid tell their neighbours). Right panel is a robustness check on the disadoption rate assumption, all curves still end above baseline. In v7 the central curve was declining, which contradicted the whole point of the report. That's fixed now. The figure validates the brief's goal of "promoting adoption of climate resilient technologies" by showing it does grow, but slowly and only under the right design conditions.

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Bibliography

Yassin:

Adil, L., Eckstein, D., Kuenzel, V. and Schäfer, L. (2026) *Climate Risk Index 2026: Who suffers most from extreme weather events?* Available at: https://germanwatch.org/en/cri [Accessed: 27 March 2026].
BANRURAL (2022) Rural microfinance and agricultural lending rates in Guatemala. Guatemala City: Banco de Desarrollo Rural. Available at: https://www.banrural.com.gt [Accessed: 27 March 2026].
Barnett, B.J., Barrett, C.B. and Skees, J.R. (2008) 'Poverty traps and index-based risk transfer products', World Development, 36(10), pp. 1766-1785. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=999399 [Accessed: 27 March 2026].
Biffis, E. and Chavez, E. (2017) 'Satellite data and machine learning for weather risk management and food security', Risk Analysis, 37(8), pp. 1508-1521. Available at: https://onlinelibrary.wiley.com/doi/10.1111/risa.12825 [Accessed: 27 March 2026].
Biffis, E., Chavez, E., Louaas, A. and Picard, P. (2022) 'Parametric insurance and technology adoption in developing countries', The Geneva Risk and Insurance Review, 47(1), pp. 7-44. Available at: https://link.springer.com/article/10.1057/s10713-020-00061-0 [Accessed: 27 March 2026].
Carter, M.R., Cheng, L. and Sarris, A. (2016) 'Where and how index insurance can boost the adoption of improved agricultural technologies', Journal of Development Economics, 118, pp. 59-71. Available at: https://www.sciencedirect.com/science/article/pii/S0304387815000991 [Accessed: 27 March 2026].
EM-DAT (2024) The international disaster database — Guatemala. Brussels: Centre for Research on the Epidemiology of Disasters. Available at: https://www.emdat.be [Accessed: 27 March 2026].
FAO (2018) Small family farms country factsheet: Guatemala. Rome: Food and Agriculture Organization of the United Nations. Available at: https://openknowledge.fao.org/server/api/core/bitstreams/1fae548a-5a22-41fd-a576-fe2aa1f77d34/content [Accessed: 27 March 2026].
FAOSTAT (2024) Crop production data — Guatemala. Rome: Food and Agriculture Organization of the United Nations. Available at: https://www.fao.org/faostat [Accessed: 27 March 2026].
Farrin, K. and Miranda, M.J. (2015) 'A heterogeneous agent model of credit-linked index insurance and farm technology adoption', Journal of Development Economics, 116, pp. 199-211. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0304387815000590 [Accessed: 27 March 2026].
INSIVUMEH (no date) National meteorological station network — Guatemala. Guatemala City: Instituto Nacional de Sismología, Vulcanología, Meteorología e Hidrología. Available at: https://www.insivumeh.gob.gt [Accessed: 27 March 2026].
MAGA (2022) Seguro Colectivo Paramétrico — programme report. Guatemala City: Ministerio de Agricultura, Ganadería y Alimentación. Available at: https://www.maga.gob.gt [Accessed: 27 March 2026].

Trocaire (2025) How climate change is driving hunger in Guatemala. Available at: https://www.trocaire.org/news/how-climate-change-is-driving-hunger-in-guatemala/ [Accessed: 27 March 2026].
WFP (2022) Indigenous women in Guatemala are insured against drought and heavy rains. Rome: World Food Programme. Available at: https://www.wfp.org/stories/indigenous-women-guatemala-are-insured-against-drought-and-heavy-rains [Accessed: 27 March 2026].
World Bank DRIFCA (2024) Disaster risk finance for Central America. Washington DC: World Bank Group. Available at: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099061925234027976 [Accessed: 27 March 2026].

BEN:
Lopez-Ridaura, S., Barba-Escoto, L., Reyna-Ramirez, C.A. *et al.* Maize intercropping in the milpa system. Diversity, extent and importance for nutritional security in the Western Highlands of Guatemala. *Sci Rep* **11**, 3696 (2021). https://doi.org/10.1038/s41598-021-82784-2
https://www.worldbank.org/en/results/2016/09/21/land-rights-indigenous-families-communities-guatemala

Wienhold, K., & Goulao, L. F. (2023). The Embedded Agroecology of Coffee Agroforestry: A Contextualized Review of Smallholder Farmers' Adoption and Resistance. *Sustainability*, *15*(8), 6827. https://doi.org/10.3390/su15086827

Mazviona B and Sølvsten S (2025) Can index insurance keep up with climate change? Rethinking historical data models. Front. Clim. 7:1649540. doi: 10.3389/fclim.2025.1649540

World bank 2024 – feasibility study report

(USDA Foreign Agricultural Service, 2016): https://apps.fas.usda.gov/newgainapi/api/report/downloadreportbyfilename?filename=Coffee%20Annual_Guatemala%20City_Guatemala_5-13-2016.pdf

Teresa
Chapter 4.1,

Abdul Latif Jameel Poverty Action Lab (J-PAL). 2024. "Leveraging index insurance to protect farmers from weather-based risk." J-PAL Policy Insights. Last modified October 2024.

Casado-Asensio, J., Kato, T. and Shin, H. (2021), Lessons on engaging with the private sector to strengthen climate resilience in Guatemala, the Philippines and Senegal, OECD Development Co-operation Working Paper no. 96, OECD Publishing, Paris.

World Bank Group. (2024) *Disaster Risk Finance and Insurance (DRFI) Solutions for family farmers in El Salvador, Guatemala, and Honduras*. Washington DC, World Bank Group.

Formatted: French (France)

Max:

Chapter 4.2:

- FAO (2018) Small Family Farming in Guatemala: Country Note. Rome: Food and Agriculture Organization of the United Nations.
- Méthot, J. (2018) 'Reconsidering non-traditional export agriculture and land in Guatemala', World Development, 110, pp. 25–39.
- NCBA CLUSA (2024) Rural Financial Inclusion (RFI) – Guatemala Project Brief. Washington, DC: National Cooperative Business Association CLUSA International.
- Root Capital (2014) Guatemala Cluster Study: Building Resilience in Smallholder Agricultural Enterprises. Cambridge, MA: Root Capital.
- Wenner, M. (2007) 'Managing credit risk in rural financial institutions in Latin America', Rural Finance Learning Centre Technical Note. Rome: FAO and IFAD.
- World Bank (2023) Guatemala Country Private Sector Diagnostic. Washington, DC: World Bank Group.
- World Bank (2024) 'Financial protection against climate risks for family farmers in Northern Central America', World Bank Blogs, 13 March. Washington, DC: World Bank Group.
- Center for Financial Inclusion (2024) Green Inclusive Finance in Action: Learnings from Guatemala. Washington, DC: Center for Financial Inclusion.
- Carter, M.R., Cheng, L. and Sarris, A. (2016) 'Where and how index insurance can boost the adoption of improved agricultural technologies', Journal of Development Economics, 118, pp. 59–71.

Chapter 4.3:

- Center for Financial Inclusion (2024) Green Inclusive Finance in Action: Learnings from Guatemala. Washington, DC: Center for Financial Inclusion.
- J-PAL (2024) Leveraging Index Insurance to Protect Farmers from Weather-Based Risk. Cambridge, MA: Abdul Latif Jameel Poverty Action Lab, Policy Insights, October 2024.
- OECD (2021) Lessons on Engaging with the Private Sector to Strengthen Climate Resilience in Guatemala. Paris: Organisation for Economic Co-operation and Development.
- WFP (2021) Climate Risk Insurance: Annual Report 2021. Rome: World Food Programme.
- World Bank (2023) Guatemala Country Private Sector Diagnostic. Washington, DC: World Bank Group.
- World Bank (2024) 'Financial protection against climate risks for family farmers in Northern Central America', World Bank Blogs, 13 March. Washington, DC: World Bank Group.

Luisa:

Belissa, T. K., Lensink, R. & Marr, A. (2025) The impact of bundling index insurance with credit and input vouchers: Experimental evidence from Ethiopia. *Journal of Economic Behavior & Organization*. 234, 107013.

Formatted: Not Highlight

Biffis, E., Chavez, E., Louaas, A. & Picard, P. (2022) Parametric insurance and technology adoption in developing countries. *The Geneva Risk and Insurance Review*. 47 (1), 7-44.

Boucher, S. R., Carter, M. R., Flatnes, J. E., Lybbert, T. J., Malacarne, J. G., Marenya, P. P. & Paul, L. A. (2024) Bundling genetic and financial technologies for more resilient and productive small-scale farmers in Africa. *The Economic Journal*. 134 (662), 2321–2350.

Bulte, E., Cecchi, F., Lensink, R., Marr, A. & van Asseldonk, M. (2020) Does bundling crop insurance with certified seeds crowd-in investments? Experimental evidence from Kenya. *Journal of Economic Behavior & Organization*. 180, 744-757.

Freudenreich, H. & Mußhoff, O. (2018) Insurance for technology adoption: An experimental evaluation of schemes and subsidies with maize farmers in Mexico. *Journal of Agricultural Economics*. 69 (1), 96-120.

Giné, X. & Yang, D. (2009) Insurance, credit, and technology adoption: Field experimental evidence from Malawi. *Journal of Development Economics*. 89 (1), 1-11.

International Finance Corporation. (2010) *Kilimo Salama – Index-based Agriculture Insurance*. Washington DC, IFC.

World Bank Group. (2024) *Disaster Risk Finance and Insurance (DRFI) Solutions for family farmers in El Salvador, Guatemala, and Honduras*. Washington DC, World Bank Group.